



ESA Annual Report 2019



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ESA Annual Report 2019

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Foreword

The global space scene is in a state of continuous evolution with, in particular, new emerging spacefaring nations, generally increased funding allocated to space activities together with an increased presence of the private sector. In this context, the European Space Agency (ESA) plays a significant role in Europe and the world, thanks to its successes with breakthrough space technologies and missions.

In my capacity as Chair of the ESA Council, I have been honoured to contribute to ESA activities in 2019 marked, in particular, by two important milestones: the ninth Space Council held on 28 May in Brussels and the ESA Council meeting at ministerial level held on 27–28 November in Seville, Space19+.

The Space Council, a joint and concomitant meeting of the Council of the European Union and of the ESA Council at ministerial level, gave wide acknowledgment of the need to coordinate actions among all of Europe's space stakeholders, as underlined by the Space Council presidencies. The Space Council is expected to discuss a common European strategic vision and goals for space, focusing on an enhanced integration of space into Europe's society and economy, and contributing to a globally competitive European space sector (Resolution on "Space as an enabler" (ESA/CE-M/CCLXXX/Res. 1(final)).

At Space19+, Member States approved a comprehensive set of activities and programmes, committing to the biggest budget to date, at €14.4 billion. The significant increase in funding of the mandatory activities, encompassing science, should be underlined, noting that all the proposed pillars attracted support: science and exploration, applications, safety and security, and enabling and support.

The present report further highlights the remarkable milestones reached in 2019, encompassing the full range of ESA activities.

Finally, I would like to welcome Ms Anna Rathsmann, who was elected Council Chair and will take up duty as of 1 July 2020.

Jean-Yves Le Gall

Council Chair

Outcome of Space19+

On 28 November 2019, ESA concluded the two-day Council meeting at ministerial level in Seville, Spain. Ministers in charge of space matters from ESA's 22 Member States plus Slovenia and Canada allocated €14.5 billion to space activities and programmes over the coming three years, including mandatory activities for 5 years, and endorsing the most ambitious plan to date for the future of ESA and the whole European space sector.

This record level of subscriptions demonstrates that ESA Member States consider space to be a strategic and attractive investment, while also recognising both its socio-economic value and ESA's role as implementer of Europe's multilateral space endeavours. ESA remains *the* European space agency, translating the investments from our Member States into tangible benefits at the local, regional, national and European levels.

The narrative for Space19+ was crafted on a foundation of four pillars: Science & Applications, Safety & Security, Applications, and Enabling & Support. Member States were asked to approve a comprehensive set of programmes to secure Europe's independent access to and use of space in the 2020s, boost Europe's growing space economy, and make breakthrough discoveries about Earth, our Solar System and the Universe beyond, as well as to strengthen efforts to secure and protect our planet. Member States assigned 35% of their total subscriptions to Science and Exploration, 33% towards Enabling and Support, 28% to Applications, and 4% to the new Safety and Security pillar.

At this meeting, the Ministers declared their support to the ESA Director General's vision "Space: The five dimensions of Space 4.0", welcoming the identification of the four programmatic pillars while inviting the Director General to further shape and promote ESA as a space agency that innovates, informs, inspires and interacts based on reliable and sustainable capabilities and infrastructure, by strengthening ESA in the worldwide context both as a leader and a reliable partner, while contributing to European autonomy in accessing, exploring and using outer space.

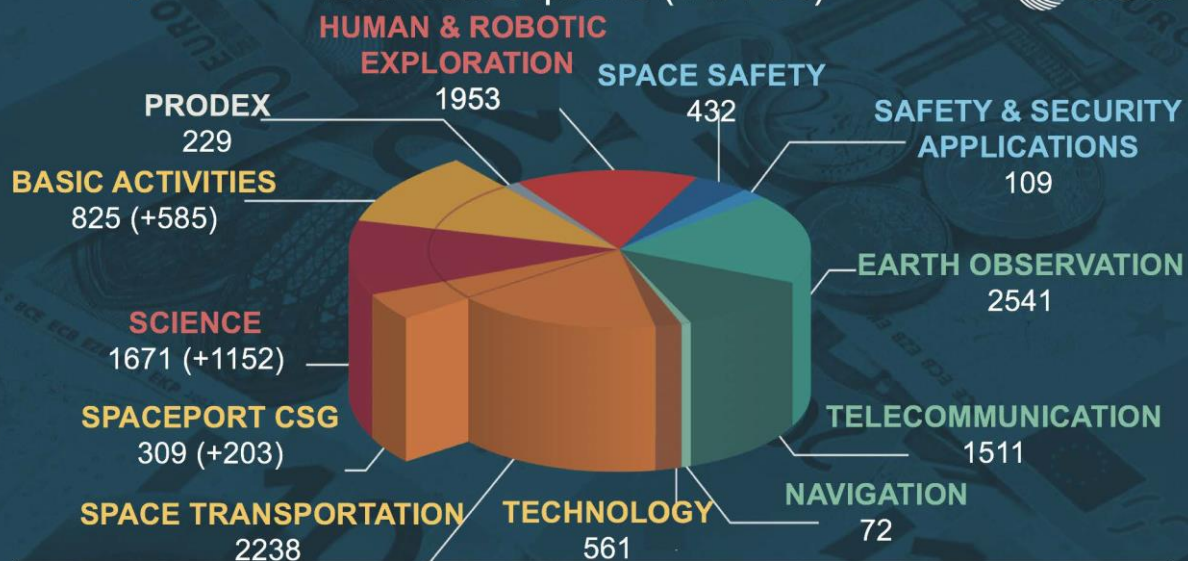
The following Resolutions were adopted at Space19+:

- Space: The five dimensions of Space 4.0
- Resolution on the Scientific Programme and the Basic Activities for Space19+
- Resolution on ESA programmes: addressing the challenges ahead

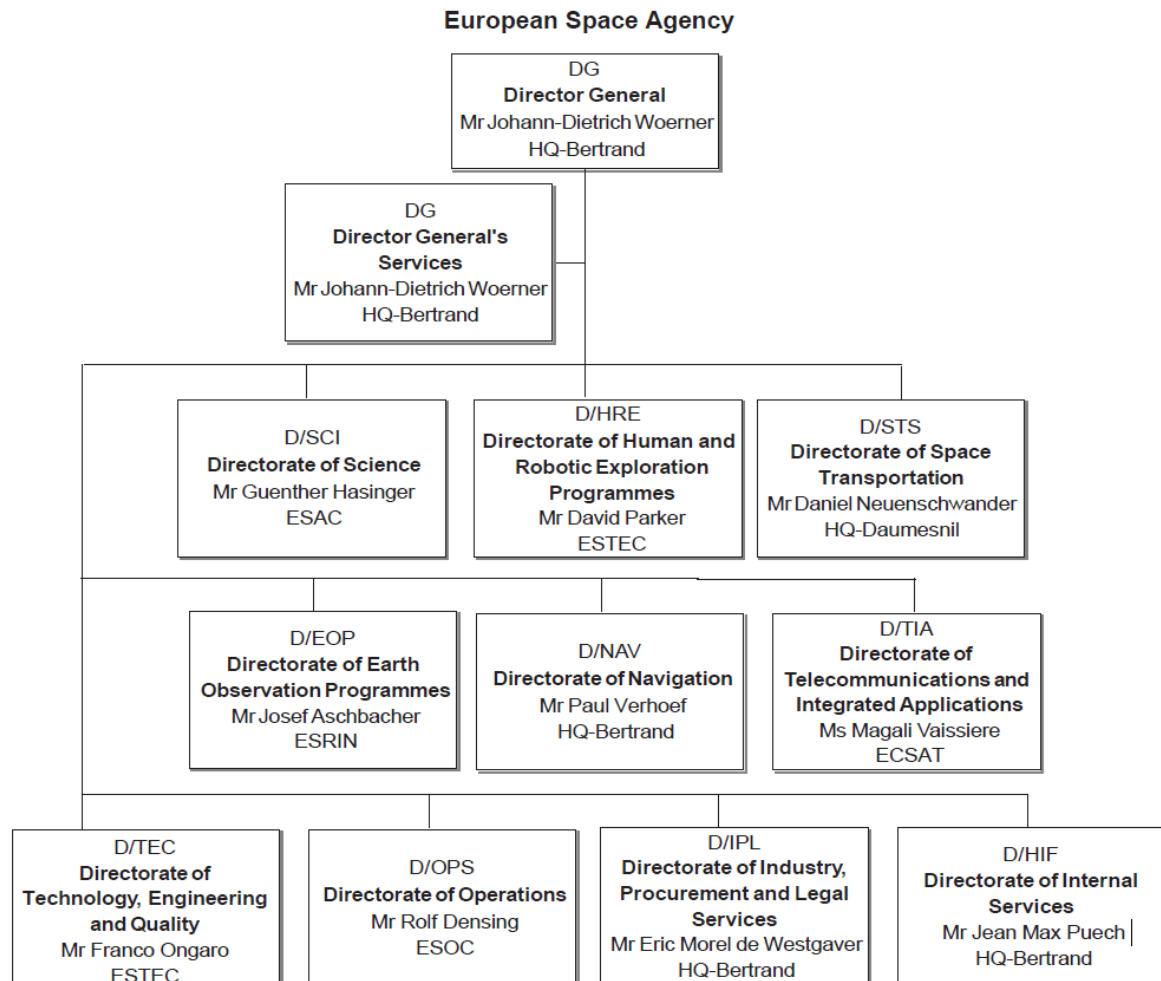
Total
Subscriptions
by
Contributor
14.4 B€

Contributor	Subscriptions in M€	Share per Contributor in %
Austria	190	1.3%
Belgium	816	5.7%
Czech Republic	150	1.0%
Denmark	128	0.9%
Estonia	9	0.1%
Finland	110	0.8%
France	2,664	18.5%
Germany	3,294	22.9%
Greece	84	0.6%
Hungary	97	0.7%
Ireland	81	0.6%
Italy	2,282	15.9%
Luxembourg	129	0.9%
Netherlands	345	2.4%
Norway	284	2.0%
Poland	166	1.2%
Portugal	102	0.7%
Romania	44	0.3%
Spain	852	5.9%
Sweden	244	1.7%
Switzerland	542	3.8%
United Kingdom	1,655	11.5%
Slovenia	5	0.0%
Canada	114	0.8%
Total Contributions	14,388	100.0%

Total Subscriptions (14.4 B€)



Management Structure



Programmes and Activities

Science & Exploration

Science

Highlights

CHEOPS

The question of whether we are alone in the Universe is perhaps one of the most profound that we can ask ourselves as a civilisation – and one that has been on the minds of scientists for centuries. The search for exoplanets – planets outside our own Solar System – received a massive boost in 1995, when a Jupiter-like planet was discovered orbiting a Sun-like star. This discovery – for which Michel Mayor and Didier Queloz were awarded a share of the 2019 Nobel Prize for Physics – marked an explosion in the field of exoplanet science. Fast-forward to today, when more than 4000 exoplanets have been found using ground-based telescopes, but especially by space missions such as Corot, Kepler and TESS. We are now moving from an era of discovery to one of characterisation. The Characterising Exoplanet Satellite, Cheops, successfully launched on 18 December 2019, will take significant steps in this direction.



Image credits: ESA - S. Corvaja - Liftoff for Cheops

Understanding exoplanets

Cheops will target individual bright stars that are already known to host exoplanets. It will use the technique of ultrahigh-precision transit photometry to accurately measure large samples of Earth- to Neptune-sized planets. By combining these measurements with known masses that have been measured from the ground, we can determine the density of the planets, and start to be able to place limits on what their internal structures and compositions could be. These limits, when combined with key parameters of the host stars and orbits, will provide vital clues as to how these small planets form, migrate and evolve.

Cheops will also be used to identify prime targets for searches for the fingerprints of key molecules by future facilities such as the James Webb Space Telescope and ESO's Extremely Large Telescope. By

already knowing where and when to point, the Cheops team will be able to monitor the exoplanet host stars before, during and after the planet transits.

Cheops was launched on a Soyuz-DS23 rocket from Europe's Spaceport in Kourou, French Guiana. It has been placed in a dusk-to-dawn Sun-synchronous Earth orbit at an altitude of 700 km and will orbit the terminator line between day and night, so that temperature variations are kept to a minimum and the photometer can be directed away from the Sun.

It is also the first small mission in ESA's Science Programme, and is implemented in cooperation with the Swiss Space Office. Smaller missions have recently been introduced to allow Member State agencies to play a greater role, while also achieving important scientific goals on a reduced budget and within a shorter timeframe. The mission's cost to ESA was less than €50m, a fraction of the cost of a typical larger mission, and was developed and implemented in five years. Cheops carries just one instrument: a high-precision telescope called a photometer, which requires a high degree of stability. The Cheops consortium consists of 11 Member States led by Switzerland and was responsible for developing the photometer.

Spacecraft design

As prime contractor for the Cheops satellite, Airbus Defence and Space – Space System Madrid (ASE) was responsible for making the platform, based on the AS-250 platform, which was designed for a bigger satellite. ASE was able to reduce the scale to fit in the limited space available on the launcher, while still providing an exceptionally stable base for the photometer.

Scientific potential

After launch, the satellite and its instrument are now operated from the Mission Operations Centre at the Spanish space agency, the National Institute of Aerospace Technology (INTA). Based in Torrejón de Ardoz near Madrid, the operations is led by ASE with support from ESA. The Science Operations Centre is at the University of Geneva, Switzerland.

Gaia

Gaia, a current flagship mission of ESA's Science Programme, reached the end of its nominal mission in July 2019, and immediately started its extended mission phase.

The mission has been designed to chart a three-dimensional map of our Galaxy, the Milky Way, in the process revealing the composition, formation and evolution of the Galaxy. The mission has been successful beyond even the most optimistic expectations and is revolutionising astronomy in many research fields. Gaia provides unprecedented positional and radial velocity measurements to produce a stereoscopic and kinematic census of over one billion stars, about 1 per cent of the Galactic stellar population.

In 2019 Gaia broke the world record for the most productive science mission ever – with over 4000 publications in a year, it surpassed the previous record holder, the Hubble Space Telescope.

A first public data release (DR) in September 2016 was followed by a much more comprehensive DR2 in April 2018. The response of the scientific community to the quality and scope of the data has been overwhelming in terms of direct feedback of enthusiasm and on social media, and an astounding rate of publications using Gaia data: 3–4 peer-reviewed scientific papers appear every day (!). The Gaia results influence practically every area of modern astrophysical research and the mission is perceived worldwide as having revolutionised astronomy.

Gaia data reveals details on future galaxy crash

The second data release from ESA's Gaia satellite has given scientists an insight into two galaxies beyond our Milky Way: Andromeda and Triangulum. One discovery is that Andromeda will one day collide with our Galaxy, but that it will be a glancing blow and 600 million years later than previously thought.



Image credit: ESA/Gaia/DPAC, CC BY-SA 3.0 IGO

This is what 150 million stars look like

This colour chart shows the distribution of 150 million stars in the Milky Way, using data from the second data release of ESA's Gaia mission, in combination with infrared and optical surveys, with orange/yellow hues indicating a greater density of stars. Most of these stars are red giants. The distribution is superimposed on an artistic top view impression of our galaxy.



Image credit: ESA/Gaia/DPAC

Human Spaceflight and Robotic Exploration

Highlights

ESA astronaut Luca Parmitano returned to the International Space Station on the Soyuz MS-13 spacecraft on 20 July. This was the beginning of *Beyond*, Luca's second spaceflight, which spans expeditions 60 and 61. During *Beyond*, more than 250 experiments took place with over 50 new experiments initiated.

Luca Parmitano became the first European to lead a spacewalk outside the ISS on 15 November. A total of three spacewalks were performed (15 and 22 November, 2 December) to upgrade the cosmic-ray-detecting Alpha Magnetic Spectrometer's (AMS-2) cooling system and extend the experiment's life in space.

At Space19+ in November 2019, Member States confirmed European support for a ground-breaking Mars Sample Return mission in cooperation with NASA. The Ministers subscribed to a total financial envelope of €1.97 billion over the next three years. With these commitments, and together with worldwide partners, Europe will take its place at the heart of space exploration. ESA will continue to commit to the International Space Station until 2030 as well as contributing vital transportation and habitation modules to the Gateway, the first station to orbit the Moon. ESA astronauts recruited in 2009 will continue to receive flight assignments until all of them have been to space for a second time, and the process of recruiting a new class to continue European exploration in low Earth orbit and beyond will be initiated.

The Maser-14 sounding rocket was launched from Kiruna (Sweden) on 24 June. The vehicle, with its 400 kg payload including two ESA physical sciences experiments: ARLES (Advanced Research on Liquid Evaporation in Space) and XRMON-GF2 (X-Ray Monitoring-Gradient Furnace2) provided more than six minutes of free-fall conditions.

The Texus-56 sounding rocket launch took place on 15 November from Kiruna (Sweden) carrying two ESA experiments: ICAPS and Perwaves. ICAPS is looking at how planets are formed from dust and Perwaves is investigating a new type of flame by burning metal powder in zero gravity.

An article about the Atmosphere–Space Interactions Monitor (ASIM) instrument was prepared and published in the prestigious *Science* magazine in its January 2020 issue. In that article, the science team reports on the discovery that extremely energetic flashes of X-rays can be emitted from thunder clouds while at the same time illuminating a huge area of the lower ionosphere by UV radiation.

The joint ESA/NASA 60-day bedrest study AGBRESA (Artificial Gravity BedRest Study with ESA) with 24 volunteers came to an end on 23 December. During the study the efficiency of two artificial gravity protocols as countermeasures were tested. In addition, seven ESA and four NASA-led experiments were part of the science programme.

Cristoforetti at helm of NEEMO 23

In the summer of 2019, ESA astronaut Samantha Cristoforetti spent time with the crew at NASA Extreme Environment Mission Operations (NEEMO), which trains astronauts for life in space. Here she is pictured with the NEEMO 23 crew, of which she is the commander, outside the Aquarius underwater habitat, located roughly 10 km off the coast of Key Largo, Florida.



Image credits: NASA–K. Shreeves

ExoMars

ExoMars rover named after DNA scientist

The ExoMars rover that will search for the building blocks of life on the Red Planet was officially named Rosalind Franklin in February. This is in recognition of the British scientist's work on the structure of DNA and also highlights this similarly ground-breaking mission's search for evidence of life on Mars.

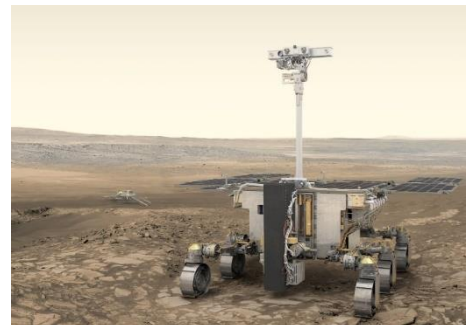


Image credit: ESA/ATG medialab

ExoMars rover ready for environment testing

Construction of the Rosalind Franklin ExoMars rover was completed and the rover was given its final touches, including its eyes. Clupi is the ExoMars close-up imager, which underwent final calibration tests at ESA's technical facility in the Netherlands. It will provide close-up views of the soil that is churned out by the drilling action.

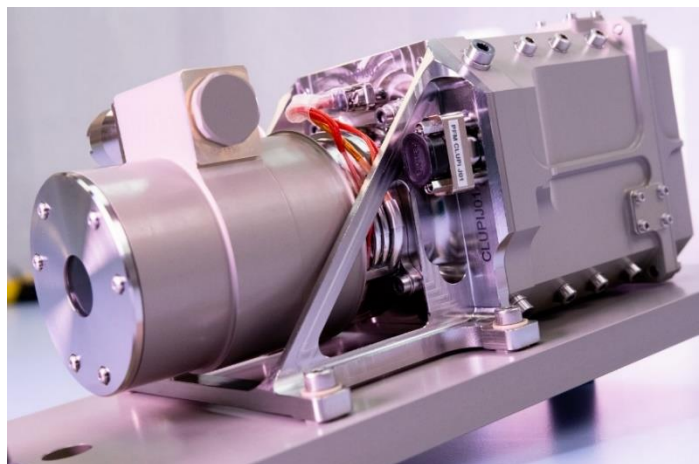
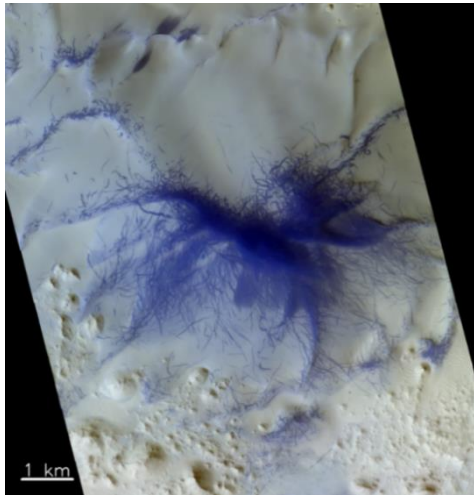


Image credits: CLUPI (ESA)

Martian landscape churned by tornadoes



A colour-enhanced image of the Red Planet taken from the ESA ExoMars Trace Gas Orbiter (a mission in cooperation with Roscosmos) shows remarkable patterns criss-crossing the Martian landscape. The marks are thought to be created by dust devil activity, which occurs when many small tornadoes converge – in this case around the crest of a ridge. The whirlwinds disturb surface dust and expose fresher material below.

Image credits: ESA/Roscosmos/CaSSIS, CC BY-SA 3.0 IGO

ExoMars data furthers understanding of Red Planet

After a year in orbit, the ExoMars Trace Gas Orbiter (TGO) has shed light on atmospheric conditions on Mars, particularly on the surprising lack of methane and the planet's huge dust storm. ExoMars TGO instruments have provided exquisite data to a much higher level than previously achieved.

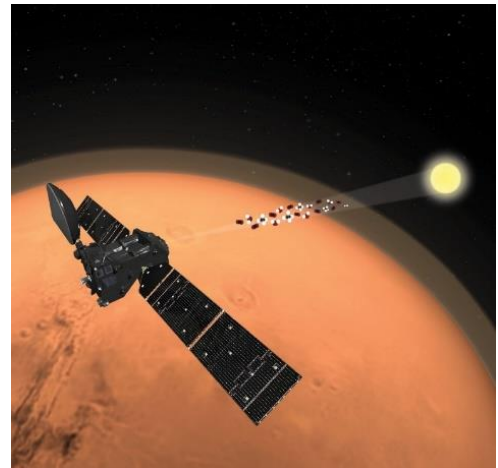


Image credits: ESA/ATG medialab

Destination Moon

As Europe prepares for future Moon missions, scientists have been studying the effects of lunar gravity to identify risks and create ways to keep astronauts healthy. ESA's space medicine team and its partners will use the planned Luna facility at the European Astronaut Centre in Cologne to help with their research.



Image credits: ESA/Foster + Partners

Powering Artemis to the Moon

The Multi-Purpose Crew Vehicle European Service Module 1 (MPCV-ESM1) is the first European contribution to the NASA-led Artemis programme while – programmatically - also serving as barter for ISS common operations costs provided by NASA to ESA. With a mass of over 15 tonnes at launch, the ESM is designed to provide the crew of Orion with propulsion, electrical power, thermal control, oxygen and water. During 2019, good progress was made on the ESM1 Assembly, Test and Launch Operations at NASA's Kennedy Space Center (KSC). The ESM was mated with the crew module, and the first combined functional tests were successful. The combined Orion vehicle will be launched on the maiden flight of the SLS on an uncrewed mission around the Moon (Artemis 1).

Meanwhile, work continued on building the second flight model ESM2 at ADS Bremen, which will carry crew into deep space for the first time since the end of the Apollo programme in 1972. The Agency also began the industrial procurement of ESM3 with a view to placing a contract in 2020, further strengthening the partnership with NASA.

Space Safety & Security

Space Safety

Space19+

At Space19+, the Ministers adopted Space Safety and Security as a new pillar of ESA activities. This will lead to new projects related to keeping the space environment operational – through the removal of dangerous debris and plans for automation of space traffic control – together with early warnings and mitigation of damage to Earth from hazards from space such as asteroids and solar flares. The Hera mission marks a joint collaboration with NASA to test asteroid deflection capabilities. New investments in the field of cyber-resilience and cyber-training were also confirmed.

Acting now to avert future asteroid impacts

Planetary defence is a key part of ESA's Space Safety Programme, with three priorities: observation, monitoring and mitigation of the risk posed by Near Earth Asteroids (NEAs). Objects wider than 15 m across can survive intact as they enter Earth's atmosphere and can cause considerable damage when they impact or if they burn up in mid-air.

Sixty-six million years ago, the Chicxulub impactor, at more than 11 km in diameter, was deadly for three-quarters of life on planet Earth. In 1908, the forest-flattening Tunguska event was most likely caused by an asteroid measuring more than 50 m across disintegrating in mid-air. Also in Russia, in 2013, the Chelyabinsk meteor, about 20 m in diameter, caused considerable damage to buildings and windows, injuring almost 1500 people.

If an asteroid the size of the Chelyabinsk or Tunguska impactors were to hit a modern city, there would be devastation. According to the ESA Planetary Defence Officer: "Having sensors that can measure and let us know beforehand would be a big advantage. Every 20–30 years something like Chelyabinsk occurs — if we have advanced warning, civil authorities could take precautions, such as evacuating the probable impact area."

Hera's mission to Didymos

Hera is ESA's contribution to the Asteroid Impact and Deflection Assessment (AIDA) collaboration with NASA. In a first step, NASA will conduct a technology demonstration mission, known as the Double Asteroid Redirection Test (DART), which will launch in 2021 and use the kinetic impactor technique to attempt to change the course of an asteroid in space by crashing a spacecraft into it.

The asteroid in question is the smaller of a double asteroid system: the 780 m diameter Didymos and its smaller satellite, dubbed 'Didymoon', which, at just 160 m in diameter, is about the size of the Great Pyramid of Giza. Both objects orbit the Sun 150–250 million km from Earth.

In the second step, Hera will launch in October 2024 and its aim is to measure DART's impact on the asteroid. It will achieve several firsts, including being the first mission to a binary asteroid system and conducting the first radar tomography of an asteroid.

According to Hera's Project Manager: "NASA's DART will be the first mission to deflect an asteroid and we hope ESA will be the first agency to assess in detail the resulting effects. The planetary defence community has reached complete consensus on the most efficient technique to use for asteroid deflection which requires demonstration: the kinetic impactor is a very simple concept – it will change the asteroid's velocity in a way that will be measurable from Earth. With DART and Hera, we will have a fully validated asteroid deflection technique enabling humanity to avoid future impacts."

Working closely with NASA on the AIDA collaboration is also an important aspect of the project: "The collaboration with another agency and its investigation team is important as this is what would happen in a real impact scenario. In addition, the Space Mission Planning Advisory Group (SMPAG), supported by the United Nations, is allowing us to put down the foundations for a wider international forum to plan future mitigation actions."

SMPAG is chaired by ESA and, together with the NASA-chaired International Asteroid Warning Network, is one of the two UN-mandated committees that lead the global effort to detect and mitigate asteroid risks.

Flyeye

The Flyeye telescope is the leading observation project within ESA's Space Safety Programme. It is due to become operational by 2021 and will be placed in a location with clear skies and little light pollution, at an altitude of 1860 m on Monte Mufara in Sicily, the same island where the first asteroid was discovered in 1801.

The innovative design of the Flyeye telescope comprises 16 lenses arranged like a fly's eye, providing an extra-wide field of vision and specifically designed for scanning rather than observation like a traditional astronomical telescope. This revolutionary telescope is just the first component in the planned Flyeye network, with plans for at least two further Flyeye telescopes in the southern hemisphere.

The Flyeye telescope project manager within ESA's Space Situational Awareness (SSA) Programme, explained: "This is an innovative project that includes several firsts, including splitting the beam of light from the incoming signal, to 16 cameras which observe different (adjacent) areas in the sky. The Flyeye will be able to detect NEOs of around 40 m in diameter, two or three weeks before they could hit the Earth, so this increases the probability of detecting an asteroid before impact. The completion of the Flyeye network would make ESA a major source of asteroid detection in the world."

NEOCC

ESA's Near-Earth Object Coordination Centre (NEOCC), at ESA's ESRIN centre in Frascati, near Rome, was building a catalogue of asteroid data in 2019. Its purpose is to evaluate and monitor the risk of NEOs and this is where Flyeye observations will be coordinated and planned. The NEOCC kept track of known NEAs, currently numbering more than 20 700, and maintained a risk list that tracks all objects for which a non-zero impact probability has been detected (currently more than 870).

Objects are listed by name and rated according to the Torino scale, which indicates in a combined scale from zero to ten the probability that the object will hit Earth together with the impact energy.

According to the Information System Manager at the NEOCC, about 160 new NEAs are discovered each month. He explains: "Most of these new discoveries are currently made by US-funded efforts – this is why Flyeye will be so important. It will enable ESA to be a key player in the discovery of new asteroids."

The impact probability is calculated according to the asteroid's diameter, its velocity and the projected time of potential impact. While larger objects, such as the Chelyabinsk or Tunguska asteroids, reach Earth far less frequently, objects greater than 15 m could still cause significant damage and are far more frequent.

Detecting the future

The future is looking bright for asteroid detection and deflection missions. As well as Flyeye and Hera, the international community is working on other innovative programmes including space telescopes, as well as new techniques, such as a gravity tractor, which entails flying a spacecraft next to an asteroid over a period of many years, with the effect of pulling the asteroid towards it due to the gravity of the two objects.

The NEOCC risk list showed that the asteroid Apophis – approximately 350 m in diameter – could potentially collide with Earth in 2068. The same asteroid is due to pass near to Earth in 2029, enabling astronomers to make very precise observations. According to the ESA Planetary Defense Officer : "After that we will know if it will impact the Earth in 2068 or not – then we will still have 40 years to plan or prepare."

Solar winds create art on Earth

In early March we joined the #AuroraHunters in Tromsø, Norway, to observe space weather at its finest: the aurora borealis. The Northern Lights appear when particles emitted by the Sun collide with Earth's atmosphere. But space weather is also potentially damaging to the infrastructure necessary for daily life and business, including communications networks, satellites and power grids.



Image credit: © Ollie Taylor. Used with permission.

How would you prepare for an asteroid strike?

In April, the [@esaoperations](https://twitter.com/esaoperations) Twitter channel covered a simulated asteroid impact exercise live on social media. It highlighted how scientists, space agencies and governments would react. In such a scenario, data from ESA's sky-scanning 'Flyeye' telescope will be crucial. This ground-breaking telescope will be located on Monte Mufara in Sicily.



Image credit: ESA/A. Baker Flyeye Observatory

Minimising our space debris

As part of ESA's Clean Space initiative, researchers took one of the densest parts of an Earth-orbiting satellite, placed it in a plasma wind tunnel then proceeded to melt it into vapour. Their goal was to better understand how satellites burn up during reentry, to minimise the risk of endangering life on the ground.



Image credit: ESA/DLR. Melting satellite part

Hera's radiation-proof brain

ESA's Hera mission to the double Didymos asteroids will rely on a fail-proof onboard computer, which has to operate up to 490 million km from Earth and withstand four years of harsh radiation exposure. The engineering model of the computer was prepared for environmental testing at QinetiQ Space's facility in Kruibeke, Belgium.

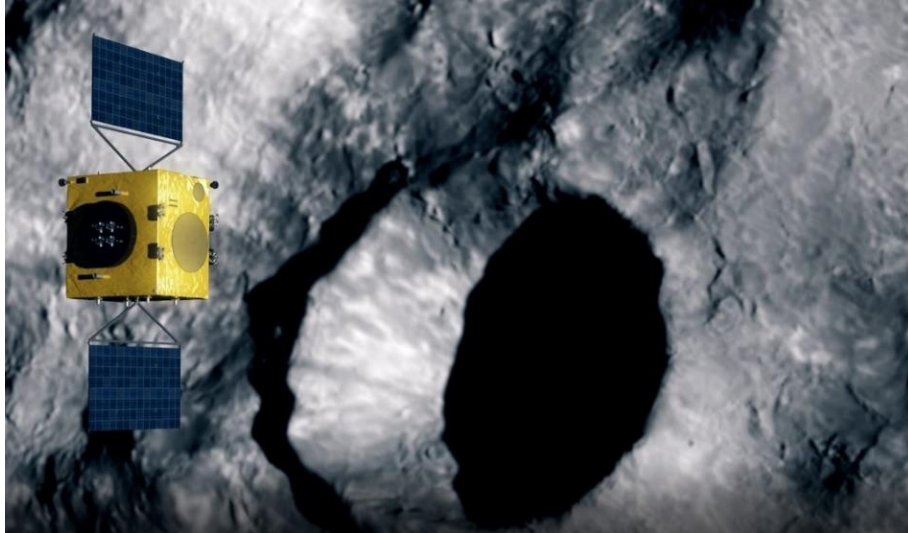


Image credit: ESA–ScienceOffice.org. Hera surveying Didymos

Planetary defence in the Eternal City

A workshop in Rome in September discussed the Asteroid Impact Deflection Assessment, or AIDA. This planetary defence mission will send NASA's DART spacecraft to crash into the smaller of the two Didymos asteroids between Earth and Mars. ESA's Hera mission will then measure the effects of the impact and how the collision alters the asteroid's trajectory.



Image credit: J. Mayor. Looking at Earth from an asteroid

Light show over the Antarctic

At the remote Concordia research station in Antarctica, the ESA-sponsored medical doctor was inspired by a beautiful display of aurora australis, or Southern Lights. In her blog she described the “green hues” and the “curtain-like waves in the sky”. Aurorae reminded us of our Sun’s activity, which will be monitored by ESA’s future Lagrange solar warning mission.

Cleaning up the orbit

The battle against space debris is key to the Agency and its Member States. The Active Debris Removal/In-Orbit Servicing (ADRIOS) project will procure the ClearSpace-1 mission to demonstrate essential technologies for guidance, navigation and control as well as rendezvous and capture methods in a world’s first active removal action. With the adoption of this new mission, ESA Member States confirmed their will to take the global lead in keeping Earth and space safe and show the way forward to essential new commercial services.



Image credit: ClearSpace SA

Cyber Resilience

Security has a pivotal role in ESA, protecting Member States' interests and reinforcing ESA's image as a reliable partner in the development of secure systems within secure environments and secure applications and services that meet the interests of Member States and the needs of the citizens of Europe. Under a new ESA Security Office managerial vision, the centralised, cross-cutting role of security and the full implementation of the ESA Security Framework has rooted all ESA activities with a solid foundation on which to build competitive reliable and secure systems and applications. Decision-making and system design following an accreditation/certification process, based on a formal initial security risk analysis, forming an integral part of the development lifecycle of each Space Programme, will ensure an adequate level of security from the programme's conception and throughout its lifecycle. ESO's Security Strategy analyses the space and security context evolution, keeping the Agency up to date in terms of security threat and risk analysis, technology competitiveness and space security application evolution in the Member State's interests.

Cyber safety and security has grown in importance due to the recognition of the criticality of space infrastructure in ensuring the availability of everyday applications and services and, in parallel, the exponential growth in innovation technologies available to potential attackers. Thus, Space19+ launched two major complementary cyber resilience activities which will be based in ESEC with functionalities deployed at ESRIN, ESOC and ESTEC and, working in synergy, will establish ESA as a leading global player in the space and security sector:

- SCCoE (Security Cyber Centre of Excellence) - which will provide a unique capability in Europe, will perform validation and testing of any space system through a synthetic environment, including of operational security procedures and critical elements, against up-to-date cyber threat scenarios. It will also represent the focal point for a Security Information Sharing capability, training and centralisation of forensic services/expertise as well as developing a distributed risk analysis process capability.
- Cyber Security Operations Centre (C-SOC), complementing the capabilities of our state-of-the-art Computer and Communications Emergency Response Team (CERT) will provide an ESA-wide cyber monitoring and management capability. The Cyber Security Operations Centre (C-SOC) will monitor and track relevant information and events with the aim of maintaining the overall Agency security posture. C-SOC will detect security incidents and support the readiness of the organisation's defensive capabilities.

Applications

Earth Observation

ESA's Earth Observation Programmes (EOP) logged a series of achievements and highlights in 2019 extending over a wide range of domains and topics. Some are discussed in the following pages.

Transversal

At Space19+, EOP obtained total funding of €2.59 billion, corresponding to 108% of what had been proposed. This constitutes the highest subscription for EOP ever. Space19+ had been prepared extensively, through numerous meetings and bilaterals. All relevant legal texts for the EOP proposals had been finalised as early as the 283rd Council meeting in October.

The Living Planet Symposium (LPS) 2019 took place in Milan from 13 to 17 May. It was the largest Earth observation event worldwide with more than 1100 presentations in 187 sessions. It attracted 4050 registered participants and 2300 students, an increase of 35% over the last LPS in Prague three years earlier, showcasing the ever increasing interest in Earth observation and its applications.

The second edition of ϕ -Week was held from 9 to 13 September at ESRIN. For reasons of available space, the number of participants had to be limited to 700, although there were 1300 pre-registrations. ϕ -Week was centred on the topic of artificial intelligence (AI). It featured a strong online component with the use of web streaming and slido to facilitate interaction. There were 12 keynote addresses during the opening, 78 talks and 160 posters, plus 23 side events. In addition there was a boot camp with entrepreneurs to develop EO prototypes.

The results of the PriceWaterhouseCoopers (PwC) study on the socio-economic benefits of Earth observation were presented to PB-EO in May 2019. As an example outcome, the study established that every euro invested in ESA's FutureEO programme will generate up to 4 euros of benefits to the European economy.

On 17 June, during the Paris Air Show Le Bourget, ESA subscribed its participation to the Space Climate Observatory (SCO) led by CNES. The first meeting of the Space Climate Observatory (SCO) Steering Committee was also held in Washington DC on 22/23 October.

The 100th meeting of the Data Operations Science and Technology Advisory Group (DOSTAG) to PB-EO was held on 3 and 4 September 2019. The Director of EOP welcomed Delegates to this special meeting and expressed his appreciation and gratitude for the work of DOSTAG.

Copernicus

Copernicus is the world's largest Earth observation programme. It is led by the European Commission, and ESA manages the space component. Copernicus features a free and open data policy and has seen massive user uptake. Seven dedicated Sentinel satellites are already in orbit. Sentinel-6, dedicated to ocean topography and slated for launch in November 2020, was shipped to IABG near Munich to begin the environmental satellite test campaign.

The Third Amendment to Copernicus Agreement was signed together with the Commission, adding €96m in funds to the Copernicus Space Component. This additional contribution is related to ESA being entrusted with further tasks such as the development of the Copernicus Sentinel-6 mission and the new European Copernicus Data Access and Information Services.

The Requests for Proposals for all six Sentinel High Priority Candidate Missions (now known as Copernicus Sentinels Expansion Missions), have been published. Based on the initial funding provided by Space19+, these missions will widen the capability spectrum of Copernicus in domains such as

climate change, food security, biodiversity, agriculture, water productivity or sea surface temperature and ice concentration. This will further enhance the usefulness and visibility of Copernicus. The project managers for the missions have also been selected.

The ESRIN Space App Camp was held from 16 to 23 September. The prize was awarded to four Norwegian developers and their app Vake|Catch, which combines Sentinel-3 data with actual data on catch amounts to help wild fisheries achieve sustainable operations.

Scientific Missions

Science remains the basis of EOP, and the Earth Explorer missions remain its fulcrum and foundation. November 2019 was the tenth anniversary in orbit of the SMOS (Soil Moisture and Ocean Salinity) mission. SMOS is ESA's first Earth Explorer mission. It has not only exceeded its planned life time, but also surpassed its original scientific goals. It was designed to deliver data on soil moisture and ocean salinity, both crucial components of Earth's water cycle. Now the SMOS portfolio has been expanded to help monitor and improve forecasting of large storms. SMOS carries a microwave radiometer to capture images of brightness temperature. Measurements correspond to radiation emitted from Earth's surface. Strong winds over oceans whip up waves and whitecaps, which, in turn, affect microwave emission from the surface. This means that the changes in radiation can be linked directly to the strength of the wind over the sea. Phase E of both SMOS and Cryosat was extended until the end of 2021 by EOEP-5 Participating States at the 177th meeting of PB-EO in February 2019.

Societal and economic benefits are also provided by ESA's other Earth observation missions. Swarm has been key for updating the World Magnetic Model which had to be provided with the magnetic north pole's new location, vital notably for navigation on smartphones. The World Magnetic Model is used to keep track of changes in the magnetic field and is updated every five years by the US National Oceanic and Atmospheric Administration and the British Geological Survey. The next update was due at the end of 2019. However, thanks in part to Swarm, researchers found that the pole is drifting in a way that was not expected. This meant that the model was too inaccurate for it to remain until the next planned revision.

The In-Orbit Commissioning Review (IOCR) for Aeolus, ESA's wind mission, was conducted at the end of January 2019, handing Aeolus over from project to mission management. The second laser of the ALADIN instrument on Aeolus was switched on at the end of June after the energy levels of the first laser had dipped to the point that the quality of the wind data was set to be compromised. The second laser works perfectly, and even better results are to be expected with the new set-up. The Aeolus team was awarded an ESA Team Achievement Award in 2019.

As for the cloud and aerosol mission EarthCARE, a project independent review of the Cloud Profiling Radar (CPR) instrument, to be delivered by JAXA, has been held. It was co-chaired by the Inspectors General of JAXA and ESA. The review resulted in a number of conclusions and recommendations that are followed up monthly via a Tracking Committee at Director level

The User Consultation Meeting on the selection of Earth Explorer 9 took place in Cambridge on 16/17 July. It resulted in a recommendation by the Advisory Committee Earth Observation (ACEO) for FORUM (Far-infrared Outgoing Radiation Understanding and Monitoring), which in turn led to a decision by PB-EO 179 in September to select FORUM for implementation as Earth Explorer 9. At a targeted overall cost of €260m and scheduled for launch in late 2025, FORUM will improve understanding of the greenhouse effect and contribute to the accuracy of climate change assessment.

Meteorological Missions

2019 saw good progress on the meteorological missions, which are being developed for EUMETSAT. Regarding MetOp Second Generation, the first Satellite System Validation Test has been held for Satellite A. As for Satellite B, the Critical Design Review (CDR) was held with the Board concluding that all review objectives have been achieved. Regarding Meteosat Third Generation (MTG), the MTG-I satellite CDR has been successfully completed.

Events and Meetings

D/EOP participated in the Space Policy Conference in Brussels on 22 January, as part of the “Space tools, Guardian Angels of Climate and Environment” session.

An ESA EOP – European Commission DG Research and Innovation Seminar was held in March in Brussels. A meeting with DG CONNECT took place on 16 April to discuss potential Science Cloud and artificial intelligence cooperation activities. A bilateral with the Commission’s Joint Research Centre took place on 17 April via videoconference to prepare for a common workshop in September.

A Sentinel-6 Pre-launch Press Event was held at IABG in Ottobrunn. Stressing the importance of international cooperation between institutions like ESA, the European Commission, EUMETSAT, NASA, NOAA and CNES, the event saw high-level attendance and raised sizeable interest among the media and general public. Sentinel-6A has been renamed Sentinel-6 Michael Freilich in the meantime.

An event entitled “Space, Security and Defence: the Next Challenges for Industry” was held at the beginning of December in Naples. It was organised by ESA, the Italian Space Agency ASI, the European Commission and the European Parliament, whose president David Sassoli was among the speakers. The event dealt with the state of affairs and evolutions in the sector and with the opportunities offered by space, also with a view to negotiations on the European Union’s next Multiannual Financial Framework (MFF).

International Cooperation in the field of Earth observation

A Copernicus Technical Operating Agreement (TOA) with Chile was signed in August. Similar agreements had already been signed earlier in 2019 with Brazil, India, Ukraine and Serbia. TOAs are non-binding agreements with non-European partners aimed at improving data access. ESA also signed an MoU with GHGSat, a Canadian SME, during φ-Week. Under this MoU, 5% of the capacity of the upcoming GHGSat-C1 (Iris) spacecraft will be offered to ESA for R&D purposes.

As for ESA’s presence in international entities, the 33rd CEOS Plenary meeting took place from 14 to 16 October 2019, hosted by the Vietnam National Space Centre. ESA will chair the Strategic Implementation Team from 2022–2023. The GEO 2019 Week took place in Canberra, Australia, from 4 to 9 November. The main themes were the Digital Economy and the Pacific Islands.

ESA also contributes to the International Charter Space and Major Disasters, which was activated 42 times in 2019.

Telecommunications and Integrated Applications

Laser precision in the night sky

ESA's Optical Ground Station in Tenerife will expand its role in laser-based communication for space missions, as well as space debris monitoring and removal. The OGS is part of the island's Teide Observatory, some 2400 m above sea level. The high altitude and clear skies make Spain's Canary Islands one of the world's best locations for astronomy.



Image credit: IAC– Daniel López Optical Ground station

A reconfigurable mission for Europe

Eutelsat Quantum is the first European satellite that can be completely reprogrammed after launch. This pioneering mission, an ESA Partnership Project with Eutelsat and Airbus as prime contractor, will influence how telecom satellites are operated in future. The satellite's coverage, frequency and power can be adjusted while in orbit. In May, the satellite's payload and platform were assembled successfully.



Image credit: SSTL Eutelsat Quantum Platform

Up and away for the SpaceDataHighway

The second satellite in the European Data Relay System (EDRS) was successfully launched on board an Ariane 5 from Europe's Spaceport in Kourou, French Guiana, on 6 August. EDRS enables data from other satellites to reach Earth faster, which can accelerate responses to emergency situations, while also opening up opportunities for the development of new services.



Image credit: ESA/CNES/Arianespace/Optique vidéo du CSG - S Martin Launch of EDRS-C

Konnect passes vibration test

Konnect, a high-throughput communications satellite, successfully completed its mechanical test at the Thales Alenia Space facilities in Cannes. This demonstrates that it can withstand the strong shaking that occurs during launch. ESA, along with the other project partners Thales Alenia Space, CNES and Eutelsat, were pleased with the results. Konnect is the first Spacebus Neo satellite.



Image credit: Thales Alenia Space. Konnect satellite in acoustic chamber

A giant leap for Europe on the worldwide telecommunications satellite market

Neosat, the ARTES programme's Next Generation Platform, has led to two successful Partnership Programmes – Spacebus Neo with Thales Alenia Space (TAS) and Eurostar Neo with Airbus – managed by ESA in cooperation with the French space agency, CNES.

It includes development and in-orbit validation of the new satellite product line for both companies, allowing the two European satellite prime integrators to deliver competitive satellites for the worldwide commercial satellite market.

Neosat has significantly reduced the cost of satellites by introducing innovation at all levels, optimising the new satellites for full electric propulsion, and shortening the time for assembly, integration and testing.

As of December 2019, Airbus and TAS have sold eleven Neosat satellites generating an exceptional return on investment. According to an independent study led by Euroconsult, for every €1 invested, more than €20 is expected to be returned to the programme's participating states and their industries.

The first satellite using the Neosat platform, Eutelsat's Konnect built by TAS on its Spacebus Neo bus, was launched in January 2020 on an Ariane 5 and marked the start of Neosat's in-orbit validation. The next one, Eutelsat Hotbird 13F based on Airbus' Eurostar Neo bus, is scheduled for launch in 2021.

Galileo Programme and Navigation-related Activities

Galileo Constellation

One billion users were benefiting from Galileo operational services by the end of 2019, an important step towards Full Operational Services. In the reporting period, Galileo system performance continued to be well beyond the specified requirements in spite of a disruption of service lasting one week in July 2019. Indeed, the Galileo satellites' ranging accuracy has been lowered to 24 cm for all satellites in the constellation, the best value ever measured for this important performance indicator. An achievement made possible thanks to the increasing number of Galileo satellites in orbit and the continuous improvement of the Ground Segment.

Additional service capabilities enabled by Galileo were successfully demonstrated with the support of ESA, contributing in particular to the service facilities integration and verification testing, important examples being the Open Service Authentication Message (OSNMA) and the Search And Rescue (SAR) return link service which will acknowledge receipt of the emergency messages.

The Constellation In Orbit Maintenance

Following the In Orbit Test (IOT) of the four Galileo satellites launched on 25 July 2018, positively concluded in January 2019, the constellation grew to include 26 satellites. An upgrade of the onboard software of the first pair of Galileo satellites launched in August 2014 was initiated in order to include these satellites, wrongly injected into elliptical orbits (due to a failure of the Soyuz rocket), in the operational constellation.

Batch 3 Satellites

The twelve Batch 3 satellites, built to maintain and complete the first generation of the full operational constellation, were under development during 2019. At the end of March, the satellite platforms were in different production islands on OHB premises. The Ground Control Segment planned to support the next Launch of Batch 3 satellites (Launch #11) under qualification to be deployed according to the L11 schedule.

July 2019 Service Incident

Migration to a new version of the ground segment, aimed at increasing robustness and security, proceeded nominally until July 2019. This was the first major upgrade since the delivery of Galileo Initial Services in 2016, including security aspects, before reaching Full Operational Capability. During this operation, a disruption of Galileo service occurred for a period of 6 days, from 12 to 17 July. Enquiry boards within ESA and the European Commission (EC) have been set up in order to investigate the root causes and identify any required measures to avoid such occurrences in the future. The resolution of this disruption incident, in full collaboration with the EC and the European GNSS Agency (GSA), is already resulting in a significant increase in the robustness and in enhanced governance of Galileo.

EGNOS

The development of the next generation of EGNOS, called EGNOS V3, which will augment the GPS and Galileo constellation and also provide double frequency messages, was kicked off at ESTEC in January 2019. EGNOS V3 will boost the performance of Galileo service over Europe, for users requiring a high level of integrity and continuity of positioning, navigation and timing signals.

A new procurement bridging EGNOS V2 and V3 (V242B) has been negotiated with TAS-F. This will allow, in particular, EGNOS V2 to improve the use of the new generations of GPS satellites for European users.

During the reporting period, the EGNOS fleet of GEO satellites has been refurbished.

Preparing the future

In 2019 ESA teams focused on the Galileo Second Generation satellite development logic.

Preparation of phase 2 of NAVISP for decision at Space19+ was also an important subject.

Concerning the High Accuracy Service (HAS), ESA undertook system engineering work to hold a High Accuracy Service Design Review (HASDR) in October 2019, supported by the GSA and EC.

Galileo Second Generation Future Satellites

The Galileo second generation (G2) programme entered a very active phase with major steps achieved:

On the one hand, G2 satellite procurement, based on the preliminary proposals from industry received in early October 2018, resulted in 2019 in the first round of competitive dialogue taking place with the three bidders, ADS, OHB and TAS. It was decided to proceed to the next phase in order to place later parallel contracts for two selected tenderers. The invitation to submit refined proposals was issued on 11 July 2019 accordingly.

On the other hand, G2 system and ground segment activities, which were based on the High Level Definition requirements, provided beforehand by the European Commission, have reached the Phase B development stage.

Research and Development in the domain of Positioning Navigation and Timing

Horizon 2020 Framework Programme for Research and Innovation in Satellite Navigation

ESA organised a GNSS Evolutions Experimental Payloads and Science Activities Call for Ideas in 2019 (ITT published in February 2019) under the Horizon 2020 Framework Programme for Research and Innovation in Satellite Navigation. The objective of the Call for Ideas was to implement a number of activities in order to develop candidate experimental payloads, new technologies for onboard atomic clocks as well as ideas for basic scientific research relevant to Galileo/EGNOS and its evolutions. The

call was open to any scientific area related to fundamental scientific research that may benefit future generations of European GNSS systems.

Navigation Innovation and Support Programme (NAVISP)

The Navigation Innovation and Support Programme (NAVISP) is composed of three different elements:

- **Element 1 – Innovation in Satellite Navigation.** The specific objectives of this Element are:
 - To perform feasibility studies and viability analysis for the emergence of new concepts in the Positioning Navigation and Timing (PNT) world;
 - To Contribute to the formulation and implementation of PNT technology innovation;
 - To evaluate the proof of concept of promising PNT-based services.
- **Element 2 – Competitiveness.** The goal of this Element is to maintain and improve the capability and competitiveness of the industry of the Participating States in the global market for Satellite Navigation, and more broadly PNT technologies and services.
- **Element 3 – Support to participating States.** This Element is focused on Participating States' priorities and aims at supporting their national navigation activities along the whole PNT value chain.

The NAVISP Programme Team has been very active and highly successful in its continued outreach effort for the Programme in the wide European PNT environment.

More than 140 people attended the first NAVISP Industry Days event organised in January 2019 at ESTEC. During the two-day event, representatives of companies and research institutions across Europe discussed dozens of projects supported by the Agency's newest programme, fostering innovation and competitiveness across the European PNT sector.

Taking stock of the programme's success, assessed by the Navigation Innovation and Support Programme Advisory Committee (NAVAC), and following recommendations of NAVAC for Phase 2, preparations for NAVISP Phase 2 for Space19+ were ongoing throughout 2019.

During Space19+, ministers delivered a clear vote of confidence to the programme by subscribing to NAVISP Phase 2 with a significantly enhanced yearly budget, thereby providing the means for the Executive to build upon the promising results achieved during Phase 1 of the programme

Enabling & Support

Space Transportation

Exploitation

Four Ariane 5, two Vega and three Soyuz launches were carried out from Europe's Spaceport in 2019, placing a total of 24 payloads of commercial and institutional customers in orbit, ranging from 7 kg to 6.6 tonnes (47 tonnes of payloads in total). European institutional payloads launched include: Prisma, COSMO-SkyMed Second Generation (CSG 1), Cheops, ANGELS, OPS-SAT and EyeSat.

Europe's launch service provider, Arianespace, reported the signature of 14 contracts with 12 customers for a total of 44 satellites in 2019. This included the contract signatures between ESA and Arianespace for the launch of institutional missions such as Jupiter Icy Moons Explorer (JUICE) and the Spanish-funded SEOSAT-Ingenio.



Image credit: ESA/P. Sebirot - Juice launcher contract signed

A Contribution Agreement between ESA and the EU on space technology activities was also signed on 16 April 2019, expanding cooperation between the two institutions. Through the agreement, the EU also contributes to ESA's Light Satellites, Low cost, Launch opportunities (LLL) initiative, specifically for the proof-of-concept demonstration flights for the Vega Small Spacecraft Mission Service (SSMS) and Ariane 6 Multi Launch System (MLS).

In November 2019, ESA, Arianespace, and industrial prime contractors ArianeGroup and Avio signed protocols on the Launchers Exploitation Phase for Ariane 6 and Vega-C. These protocols will govern the long-term exploitation of these launch vehicles. They cover aspects related to technical and industrial responsibilities in the wide range of areas pertaining to operations such as compliance with high-level requirements over the lifetime of both vehicles, launch authorisation, configuration management, and maintenance of various assets.

Ariane

2019 marked the 40th anniversary of Ariane since the first Ariane 1 launch on 24 December 1979. Incidentally, the last Ariane 5 flight of 2019 was the 250th flight of an Ariane from Europe's Spaceport.

2019 also saw the launch of EDRS-C (European Data Relay System) on Ariane 5 flight VA249. On that launch, Ariane 5 featured new slightly elongated upper-stage cryogenic tanks. This made it possible to carry several hundred kilograms more propellant, bringing a launcher performance gain of about 100 kg payload allowance to geostationary transfer orbit, corresponding to an extended burn duration of about 30 seconds.

A new Ariane 5 configuration with extended tanks in the upper stage and a new lighter Vehicle Equipment Bay VEB-H was qualified in 2019. The first flight with this new configuration was scheduled to fly in 2020.



Image credit: John Kraus – Launch of VA250

Vega

On 22 March, Vega's 14th mission took place from Europe's Spaceport in French Guiana. It manoeuvred into its target Sun-synchronous orbit about 54 minutes into the mission after two burns of its upper stage. Complying with debris regulations to help keep space clean, Vega's upper stage fired a final time to ensure direct reentry and burn up high in the atmosphere over the ocean.

On 10 July, approximately two minutes after liftoff, Vega flight VV15 suffered an anomaly that led to the failure of the mission. An Independent Inquiry Commission (IIC) co-chaired by ESA and Arianespace representatives identified the anomaly's most likely cause: a thermo-structural failure in the forward dome area of the Zefiro-23 motor. Based on the recommendations of the IIC, ESA and its industrial partners took corrective actions to allow Vega launches to resume in 2020.

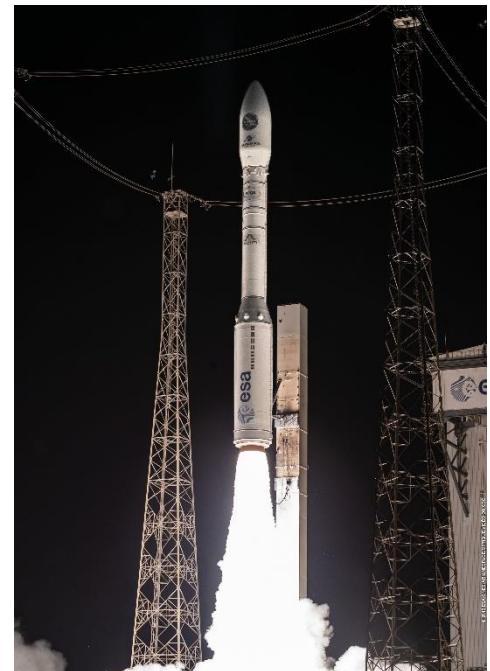


Image credit: Arianespace – Launch of Vega VV14

Development and future preparation

Ariane 6

Launcher System

The final qualification static firing test of Vulcain 2.1 took place in July 2019. The test lasted 655 seconds. This completed a total of 13 798 seconds of operation, or nearly four hours with a controlled engine, using Ariane 6 flight actuators to gimbal the engine. A review in September 2019 marked the culmination of two Vulcain static firing test campaigns over 15 months on two demonstration models in test facilities at the DLR test centre in Lampoldshausen, Germany. The refurbished engine was subsequently delivered to IABG, also in Germany, for mechanical tests.

A new test facility called P5.2 was inaugurated in February 2019 at the DLR Lampoldshausen test centre. This facility will test the complete Ariane 6 upper stage in conditions as close as possible to those experienced in flight. The tests will include Vinci engine firing, non-propulsive ballistic phases, tank pressurisation to increase performance, Vinci reignitions, exhaust nozzle manoeuvres, and passivation.



Image credit: ESA/S. Corvaya – Ariane 6 upper stage 'Test cell'

During the summer of 2019, a first simulation with Ariane 6 software was performed on the separation of spacecraft with a Galileo mission. This complex mission made it possible, in particular, to verify the correct behaviour of the Guidance, Navigation & Control algorithms.

Launch Base

A number of Ariane 6 Launch Base reviews were conducted in 2019:

- The Steering Board of the Launch Base Critical Design Review (CDR) was successfully held on 12 June 2019.
- A review focusing on Ariane 6's launch zone (ELA-4), with a special focus on adaptations of the Launch Range systems that ensure safety.
- An ELA-4 Infrastructure Test Review Board was successfully held on 16 and 17 July 2019.

During July 2019, the first rollout of the Ariane 6 mobile gantry took place, followed by a series of tests that mimic pre-launch operations. When fully equipped, this mobile gantry will weigh 8200 tons. Its platforms provide access to the launch vehicle for integration on the launch pad. It protects Ariane 6 until its doors are opened and is retracted about five hours before the launch.

The mobile gantry stands on 16 bogies, each comprising eight wheels and with each wheel equipped with one electric motor. A total of 128 electric motors synchronise to set the wheels in motion along rails.



Image credit: ESA – Ariane 6 mobile gantry over the launch pad

The Launch Base Technical Qualification mechanical test campaign was performed from 5 to 24 August 2019 followed by the second part of the campaign in December 2019 to validate the equipment and procedures used to erect the Ariane 6 central core at the launch pad.

After a successful Launch Base Technical Qualification Review, CNES will hand over the launch base to ESA.

Launch System

The Launch System Critical Design Review was closed on 16 July 2020, and contracts finalised with ArianeGroup and the French space agency (CNES) for the preparation and execution of combined tests.

Vega C

The Vega C Launch System Critical Design Review took place on 20 February 2019. Independent experts from ESA, national institutions and industry confirmed the consistency and maturity of the entire Vega-C launch system design, authorising the programme to enter the design qualification phase. The review included all the elements of the Vega-C launch system, as well as integration activities for the launch base and launch range.

The second static firing test of Vega-C second stage (the Zefiro-40 motor) took place in May 2019, alongside casting of the Zefiro-40 Flight Model 1 (FM1).

Subsequently, in December 2019, a final definition was decided upon of both the Vega-C configuration and high-level schedule.

Building works for the new remote Vega Control Centre at Europe's Spaceport are progressing with the objective of completing the building in 2020.



Image credit: ESA/J.Huart – Artist view of Vega-C on the launch pad

P120C

The static firing test of the first qualification model of the P120C (QM1) solid rocket motor was completed at Europe's Spaceport on 28 January 2019. Fully loaded with 142 tonnes of fuel, the 13.5 m long, 3.4 m diameter motor was ignited for a simulation of liftoff and the first phase of flight in the Vega-C configuration.

During a burn time of 135 seconds, the P120C delivered a maximum thrust of about 4650 kN. No anomalies were seen and, according to initial recorded data, the performance met expectations.

This was the second P120C static firing test, following the test of the P120C development model in July 2018. A third test, for the second qualification model (QM2), in Ariane 6 configuration, will be tested in 2020.



Image credit: ESA – Hot firing of P120C solid rocket motor for Vega-C

Space Rider

With Space Rider, Europe will enter a new era through the demonstration of operational transportation capabilities for in-space operations and for return from space, thereby enabling European industry to open up new markets. It will be the first European reusable space transportation system, offering Europe an operational reentry and landing capacity based on a multi-purpose uncrewed free-flyer platform. It will be launched into low Earth orbit on Vega-C and remain there for at least two months at a time. Space Rider will carry experiments that will contribute to research for industrial applications such as materials, pharmaceuticals, Earth observation and technology demonstration. At the end of its mission, Space Rider will reenter our atmosphere and land safely back on Earth, before being refurbished and launched again. Space Rider's Phase C (Detailed Definition) started in January 2019. The subsystem Critical Design Review process closed later during the year.



Image credit: ESA/J.Huart – Artist view of Space Rider

Future Launchers Preparatory Programme (FLPP)

Prometheus

The Prometheus engine is a precursor for a new liquid rocket engine family that has been designed for low cost, flexibility, and reusability.

The project aims to design, produce, and test an advanced low-cost 100-tonne-class LOx/LCH₄ reusable engine. The engine is designed for €1m of recurrent costs and targets flexibility in operation through variable thrust, multiple ignitions, compatibility with main and upper stage operation, and minimised ground operations before and after flight.

Production features additive manufacturing for the production of engine components, the introduction of a full electric command system, and the onboard rocket engine computer (REEC) for engine management and monitoring.

The manufacturing of the first Prometheus test engine started in April 2019. The last test campaign of the Gas Generator version 3.0 at the P8 test bench was finalised at the end of October 2019.

Modifications are being made to the P5 test bench at the DLR test facility in Lampoldshausen in order to test Prometheus. The Critical Design Review of these modifications was performed in November 2019. A review of the preliminary design of the methane supply system was held the same month.



©ArianeGroup Holding

*Image credit: ArianeGroup Holding –
Rendering of Prometheus engine*



*Image credit: ArianeGroup – Expander-
cycle Technology Integrated Demonstrator*

Expander-cycle Technology Integrated Demonstrator

The Expander-cycle Technology Integrated Demonstrator, or ETID, is a full-scale integrated demonstrator for an upper-stage rocket engine. ETID is a precursor of the next generation of 10-tonne rocket engines. Results from the project should pave the way towards a lightweight, high-performance rocket stage, while also indicating methods of substantially reducing its serial production cost and thereby honing its market competitiveness.

The ETID full-scale engine demonstrator test campaign was finalised in 2019 leading to a review post tests at the end of May of the same year. The test campaign lasted 9 months and took place on the P3.2 test bench at DLR Lampoldshausen. Overall, 23 tests were performed for a total operating time of 2707s. During the testing, 49 different operating points were reached, including testing the behaviour in “extreme” regimes – such as increasing the flow of cool hydrogen in the system and therefore “over-cooling” the hardware during operation. This shows the versatility of the ETID design to operate over wide mixture ratio and chamber pressure ranges.

In the first step towards an ‘intelligent’ engine, there is a fault-tolerant controller associated with the valves. If any electrical component fails, the control system automatically compensates so there is no functional impact.

Storable Propulsion Demonstrator

The Berta engine to demonstrate storable environmentally-friendly propulsion, achieved a System Requirements Review of its throttleable injector.

Microlauncher technologies

ESA also selected microlauncher underlying critical technologies to be further matured within ESA's existing technology portfolio, also bringing the prospect of validating and embarking technologies on large-scale demonstrators. The first batch of technologies relate to low-cost avionics, composite tanks, the separation system, turbopumps and the safety framework for microlaunchers.

Microlauncher technologies are maturing, as demonstrated during a successful drop test of a reusable first stage performed by Spain's PLD Space on 11 April 2019. This drop test was carried out from the El Arenosillo Experimentation Centre in Spain.



*Image credit: PLD Space –
PLD reusability drop test*

Phoebus Demonstrator

In May 2019, MT Aerospace and ArianeGroup signed contracts with ESA to develop “Phoebus”, a Prototype of a Highly OptimisEd Black Upper Stage.

The low-cost lightweight Phoebus demonstrator introduces carbon composite materials, in particular for the tanks containing the cryogenic propellants such as liquid hydrogen and oxygen, and for other primary and secondary structures.

The goal is to achieve optimum mass savings while ensuring propellant compatibility with these new materials, at a lower production cost.



Image credit: ArianeGroup – Phoebus

Commercial Space Transportation services and support to Member States

Boost!, the new programme on Commercial space transportation services, was started following adoption at Space19+. This programme is born out of ESA's long-term vision to support private economic operators in their undertakings to develop new European space transportation services. Boost! will co-fund and assist the pre-commercial development of new European space transportation services ranging from launch opportunities, in-space transport services to return from space capabilities.



Image credit: CNES/ESA/Arianespace/Optique Vidéo CSG/P Baudon, 2; John Kraus – Ariane 5 liftoff

Europe's Spaceport – Guiana Space Centre

At Space19+, the CSG Resolution (2020–2024), including investments for the Core Launch Range Renewal, was approved.

The Core Launch Range Renewal (CLRR) activities are a series of investments in the modernisation and improvement of Europe's Spaceport. The focus of these investments is on increased flexibility and cost reduction in the operation of the facilities. This could eventually translate into a reduction of the overall launch costs and thus an increase in competitiveness. These investments are also intended to contribute to energy transition by reducing the CO₂ footprint and energy consumption with the use of a smart grid and a shift towards renewable energy.

In order for these investments to be effective, a fundamental restructuring will take place for the spaceport workflows and procedures. New infrastructure will be built and personnel will be trained on new working methods. Automation, remote monitoring and control are technologies at the core of this modernisation. Some investments will also cover needed end-of-life overhauls for systems such as road networks, water distribution and air conditioning.



Image credit: ESA/S. Corvaja– Vega and Ariane 5 launch pads at Europe's Spaceport

Operations

High-speed connectivity to outer space

ESA's Mission Operations Infrastructure improved the data throughput of its deep space network by a factor of 15 through a major 26 GHz band upgrade. It ensures a reliable high-speed data downlink for the next decade supporting missions such as Euclid. The Malargüe station in Argentina was the last of the stations to be upgraded at the end of 2019.



Image credit: ESA/D. Pazos - Comet relay

OPS-SAT – Flying Advanced Missions Operations Concepts

18 Dec 2019 marked the launch of OPS-SAT, ESA's first cubesat dedicated to operations and fully developed and operated by ESOC. The satellite was developed by a European consortium under GSTP funding. It will study advanced mission operations concepts on a flying test-bed. One of the many experiments is paving the way for interchangeable apps in space.



Image credit: ESA - Artist's impression of Ops-Sat

Technical and Quality Management

State-of-the art bioprinted skin and bone samples were produced as part of an ESA Discovery and Preparation project, investigating the 3D printing of human tissue to help keep astronauts healthy all the way to Mars. The samples represent the first steps in an ambitious end-to-end roadmap to make 3D bioprinting practical for space.



Image credit: ESA/SJM Photography - Bioprinted bone sample

A new CubeSat Systems Unit was set up as a centre of excellence, building up deep expertise in miniaturised technology, equipment and systems integration, at the service of CubeSat projects across the Agency. The Unit works to tailor existing European Coordination for Space Standardisation regulations specifically for CubeSats, and facilitating access to CubeSat-friendly test facilities.

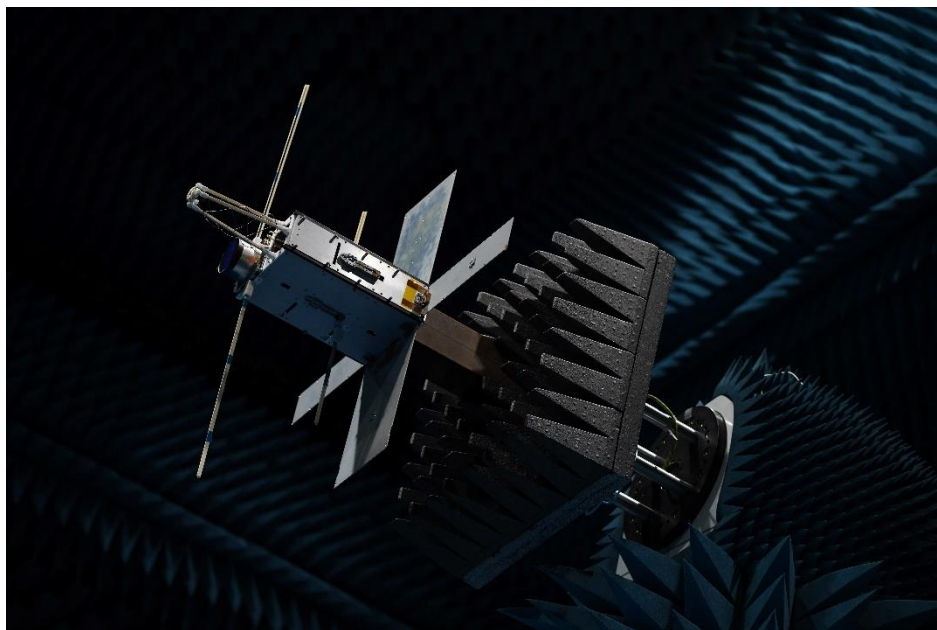


Image credit: ESA/G. Porter - Hiber CubeSat testing

The so-called ICE Cube was implemented by ESA (TEC), leveraging the platform developed by the Belgian company Space Applications Services. The instruments inside the cube were provided by ESA (TEC) and CERN. The Cryptography ICE Cube experiment aims to validate two distinct technologies for securing communications in space without the need to use radiation-hardened components. This is especially useful for small or experimental missions like CubeSats. ICE Cube was launched and commissioned in April 2019 and has been returning useful data ever since.

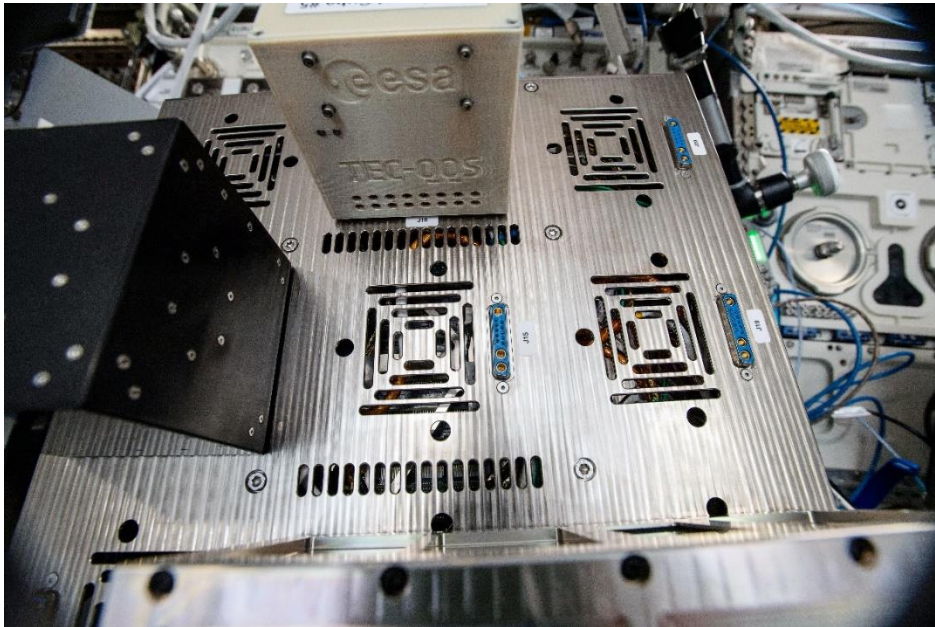


Image credit: ESA/G. Porter - Cryptography ICE Cube experiment

One such facility, ESTEC's Mechanical Systems Lab – a small-scale version of the adjacent Test Centre, serving spacecraft instruments, subsystems or mini-satellites rather than full-size missions – spent July 2019 establishing the spaceworthiness of ESA's Lander Radioscience experiment, a milk-carton-sized payload destined for the ExoMars Surface Platform in 2022.



Image credit: ESA/M. Cowan - LaRa team beside thermal vacuum chamber

In May 2019 Franco Ongaro, Director of Technology, Engineering and Quality, and Rolf Densing, Director of Operations, in the presence of Günther Hasinger, Director of Science, signed an agreement with the Instituto de Astrofísica de Canarias (IAC) on laser-based activities in the clear skies of the Canary Islands. This agreed the extension and expansion of ESA's existing activities in Tenerife as an 'anchor site' to install optical ground stations for operational two-way satellite to ground links and for developing laser-based space debris removal techniques.



Image credit: ESA/D. López - ESA's Optical Ground Station

More than 8000 members of the public visited ESTEC in October, for the eighth annual ESA Open Day in the Netherlands. With the theme of ESA to the Moon, to mark the fiftieth anniversary of the first lunar landing, people from all over Europe met astronauts, space experts and saw behind the scenes of Europe's space adventure at ESA's largest establishment. For the occasion, the ESA astronaut of Dutch origin André Kuipers was joined by Apollo astronauts Walt Cunningham and Rusty Schweickart.



Image credit: ESA - ESA Open Day in the Netherlands 2019

In November, ESA's Human Robot Interaction Laboratory helped ESA astronaut Luca Parmitano make robotics history in the conclusion of the Analog-1 project. From the International Space Station, Luca controlled an Earth-based rover, equipped with an advanced gripper possessing the equivalent mobility and dexterity of a human hand, to perform a two-hour geological test campaign in a mock-lunar environment.

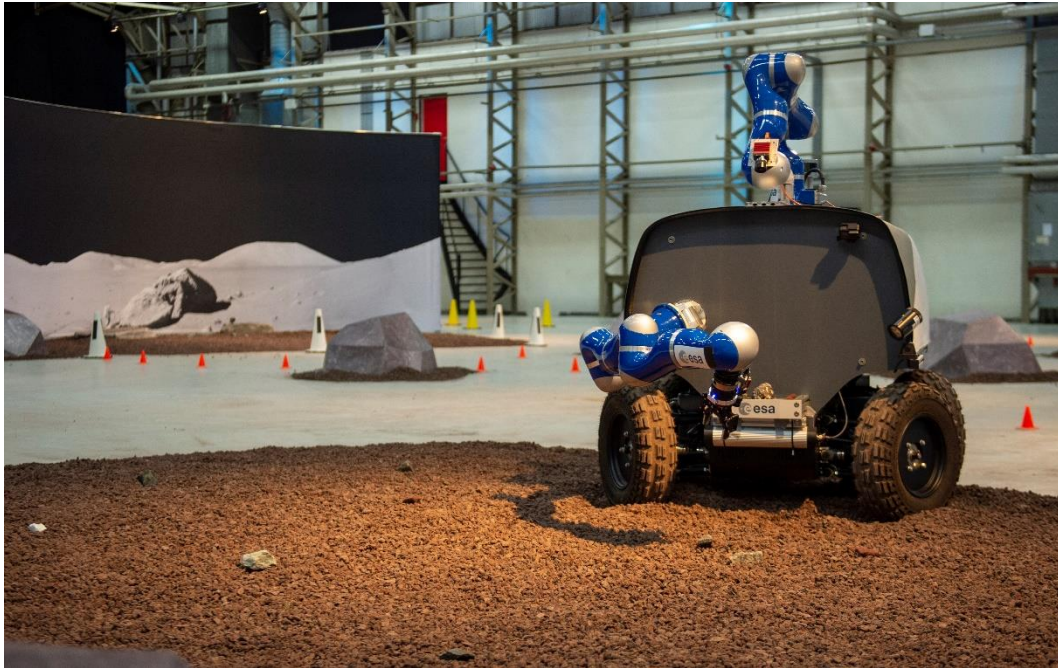


Image credit: ESA/G. Porter – Analog-1

The same month, the Space19+ ministerial gathering setting the Agency's future course saw heightened funding of R&D programmes and a strong endorsement of the Directorate's new Technology Strategy, together with agreement of the Directorate-developed Hera asteroid mission for planetary defence.



Image credit: ESA/ScienceOffice.org - Testing deflection

2019 marked another busy year for the ESTEC Test Centre – at 3000 square metres in size, the largest satellite test facility in Europe – with test campaigns including the 10.5-m long deployable magnetometer arm for ESA’s Jupiter mission Juice, MetOp-SG Structural and Thermal Model and the 10-m high protective fairing of the Vega-C launcher. The Test Centre ended the year by taking delivery of the first in the next batch of Galileo satellites.



Image credit: ESA/G. Porter – Juice’s magnetometer boom



Image credit: ESA- Vega-C fairing’s trial by sound

In April, ESA harnessed a new resource for future space activities, when it opened a new platform inviting anyone – from researchers to the general public – to respond to space-related challenges. The Agency’s new Open Space Innovation Platform (OSIP) website forms part of a wider effort to support the future competitiveness of Europe’s space industry with an early technology development strategy. In November, ESA opened the Open Discovery Ideas Channel, and the first 50 ideas are being implemented, kick-starting a new era of space research.

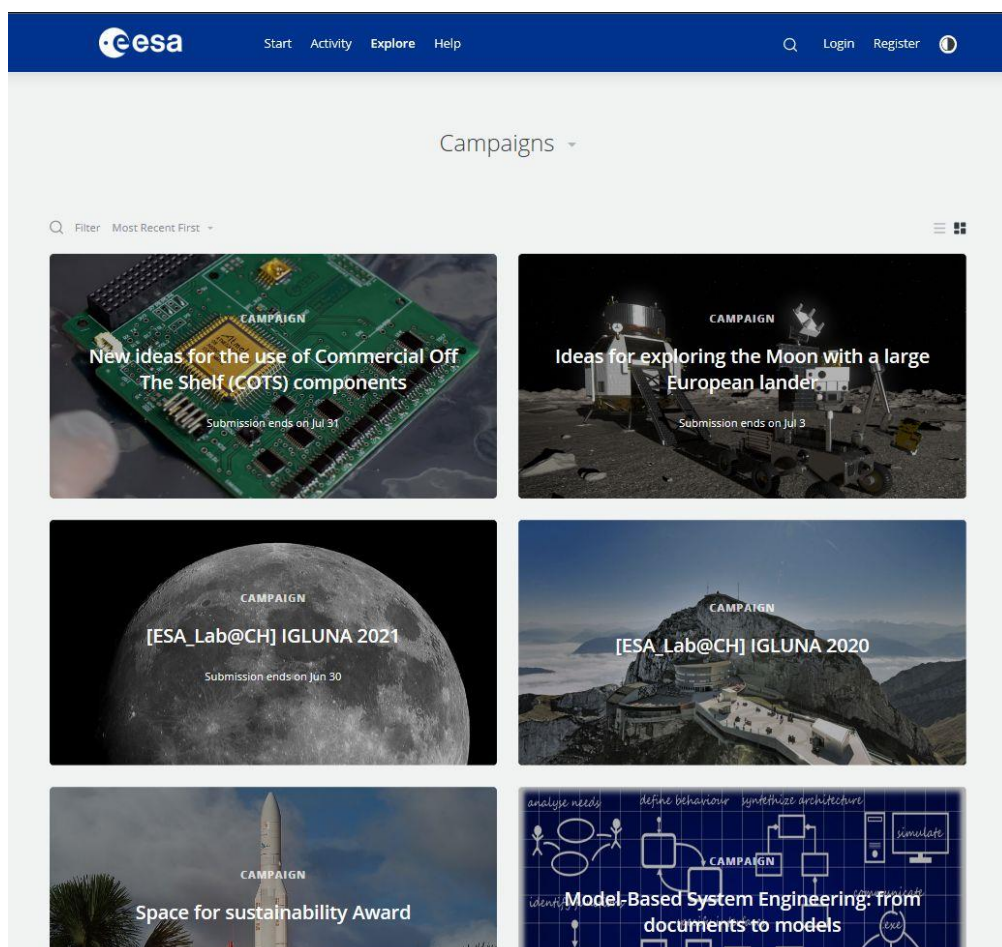


Image credit: ESA-Open Space Innovation Platform Campaigns

ESA took the first steps in 2019 towards putting in place a dedicated strategy for using more “Commercial off the Shelf” (COTS) elements in ESA spacecraft by organising together with *Alter Technology* and the *University of Seville* a dedicated workshop called “ACCEDE” (Assessment of Commercial Components Enabling Disruptive Space Electronics). The workshop was attended by 250 international experts, including satellite providers, operators, component manufacturers and even insurance underwriters. The outcome from the workshop forms a basis for an ESA 11-point workplan on future COTS usage, which will be implemented between 2020 and 2021.

Following the launch of OSIP, the GSTP initiated the Digital Technologies for Space Call for Ideas. Over 150 proposals were received through the platform on themes ranging from artificial intelligence, to cybersafety and additive manufacturing. These ideas were then discussed openly by more than 100 non-traditional space players at July’s Digital Technologies Working Days event in Estonia, where participants tried to find solutions to the most pressing challenges being faced under each topic.



Image credit: ESA - Merle Maigre, CybExer Technologies

As of 2019, every €102,000 worth of General Support Technology Programme (GSTP) contracts has led to one additional job and that every €1 invested in product development through the Programme results, on average, in €3.5 of additional commercial benefits. In 2019, the GSTP initiated 158 new activities, ranging from developing demisable propellant tanks, to the PRETTY (Passive Reflectometry and Dosimetry) CubeSat that will measure sea ice for climate change monitoring. 2019 also saw the Programme undergo an evolution to meet the goals of Space19+ through targeted frameworks and initiatives, such as a knowledge-sharing enterprise, which will expand additive manufacturing awareness and experience across small- and medium-sized companies in several Member States.

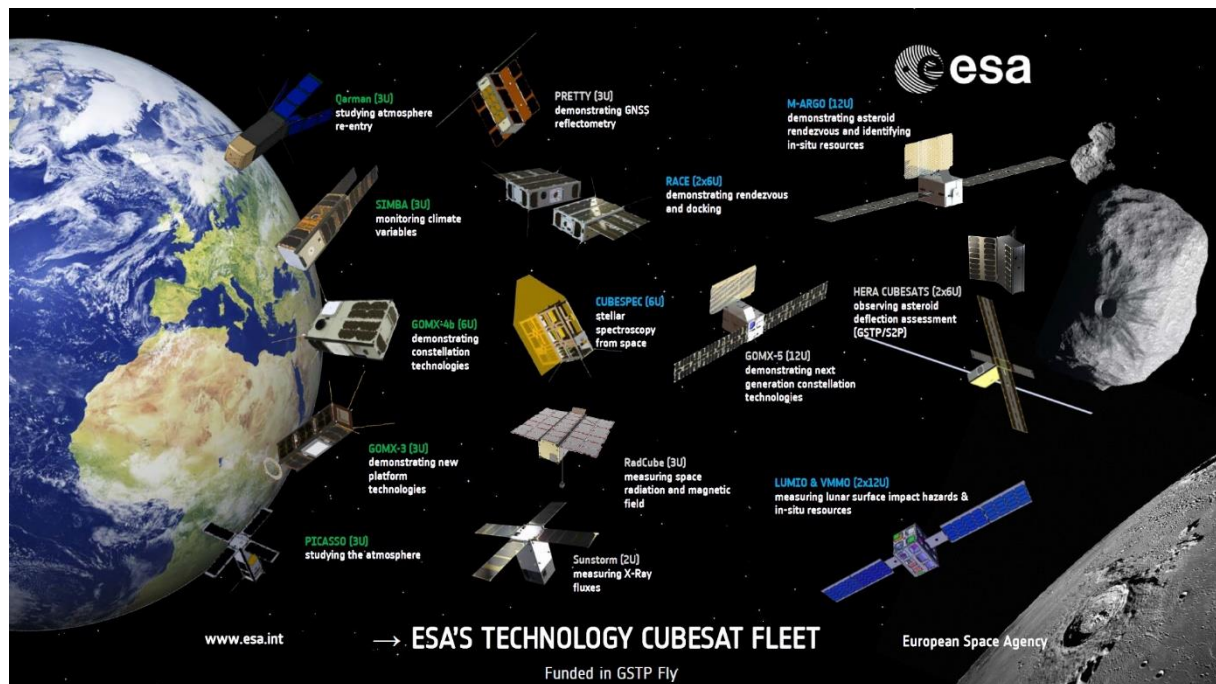


Image credit: ESA - ESA's Technology CubeSats

Between January and December 2019 the Technology Development Element (TDE) initiated 125 new activities and closed 135. One of these is a new refuelling mechanism, ASSIST, which will allow satellites to be refuelled and serviced while already flying in orbit around Earth, hopefully extending their lifetime. This activity designed a servicing and refuelling system for GEO satellites that can be standardised across the space industry, allowing all future satellites to be equipped with basic interfaces for capture and refuelling.

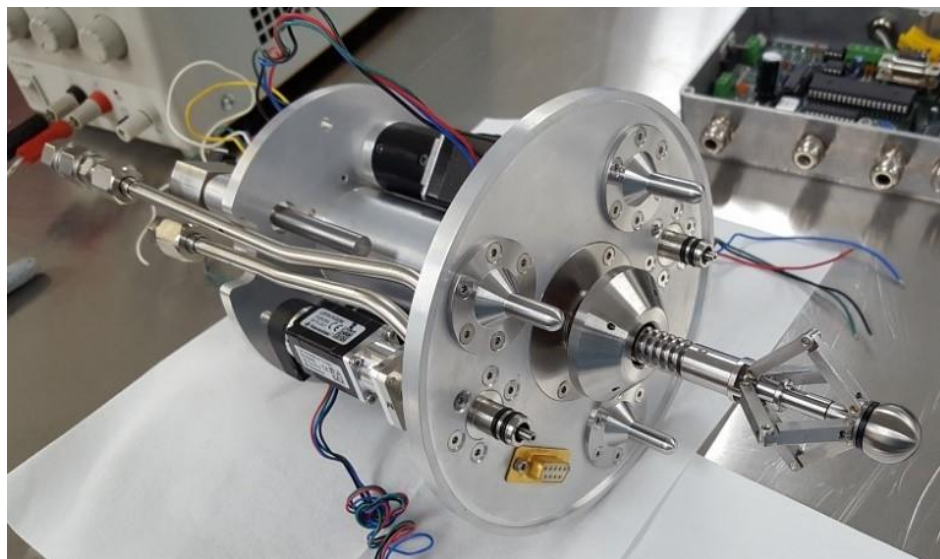


Image credit: ESA - ESA's Technology CubeSats

On 18 December, 2019, OPS-SAT was injected into a circular, polar orbit at 515 km altitude. The first of its kind, OPS-SAT is an open, flying 'laboratory' available for in-orbit demonstration of revolutionary new control systems and software that would be too risky to trial elsewhere. Over 100 companies and institutions from 17 European countries registered experimental proposals to fly the mission. The GSTP-funded spacecraft was launched as a 'secondary payload' along with the CHEOPS planet-characterising mission, for which GSTP integrated all of the AIT support activities and supports the mission operations centre.



Image credit: ESA/ S. Corvaja- Liftoff for Cheops

The Helicon Plasma Thruster (HPT) is a concept of electric propulsion which generates thrust by electrodeless acceleration of plasma. This propulsion technology can offer a good level of performance while eliminating many of the design and manufacturability issues which have afflicted electric propulsion systems to date (electrodes, high voltage electronics, complex fabrication), therefore proposing a cost-effective solution for large constellations of small satellites. The ESA Propulsion Laboratory (EPL) has supported the development of this technology since its first prototype, back in 2015, which demonstrated the feasibility of the concept by igniting and accelerating a flux of argon with an external RF generator. Successive iterations of the HPT prototype have been tested in additional campaigns at the EPL and on industry premises, achieving improved levels of ionisation and acceleration efficiency in the thruster at the same time as the mechanical and thermal design evolved into a compact, lighter iteration. The success of this process was demonstrated when this technology was selected for a grant under the European Commission Strategic Research Cluster on Electric Propulsion, with the aim of developing and verifying HPT technology for application in non-geostationary satellite constellations and other small spacecraft.

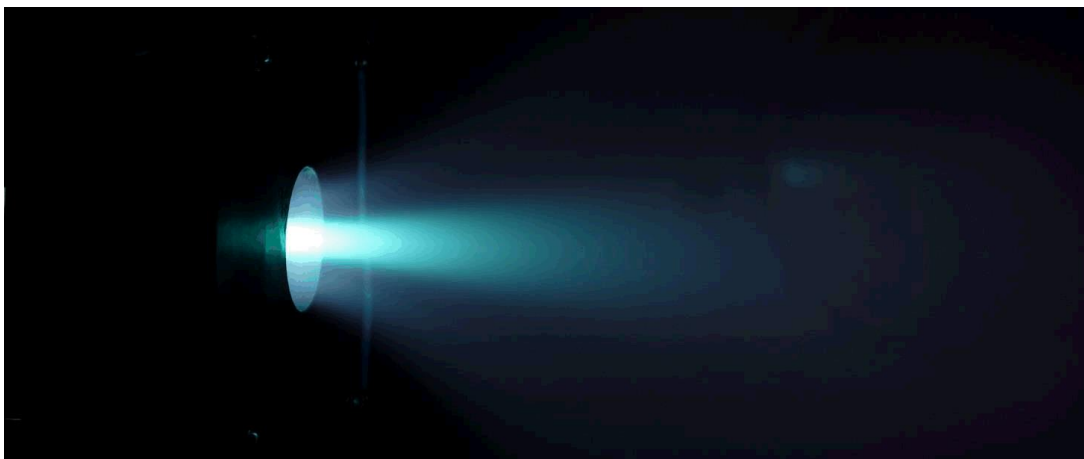


Image credit: SENER - SENER Helicon Antenna Thruster firing at the ESA PROPULSION LAB (Owner: Sener)

A major highlight of the year was the completion of the Solar Orbiter thermal verification that was brought forward in a well-defined combination of TB/TV tests at unit/subsystem level, tests at heat shield level at high flux intensities (about 13 solar constants) and tests at spacecraft level with intensities of up to 1800 W/m² (1.3 solar constants). The last missing part was the thermal verification of the complete Solar Array Assembly. The solar array panels were fully tested but later in the course of the project there was a need to add an electrically conductive tent to the reverse side of the panels and these tents were tested separately in thermal cycling tests but without vacuum. To complete the verification, another coupon was manufactured and subjected to thermal cycling test in vacuum, pull test on the fixation items and grounding resistance test. All requirements applicable for the electrical, mechanical and thermal vacuum cycling tests on the Solar Orbiter Confidence Coupons were met at each test step. After the completion of the tests, Solar Orbiter was shipped to the Kennedy Space Center (KSC) for final integration and tests before launch. From the thermal point of view, only minor multi-layer insulation adjustments were needed.



Image credit: NASA/B. Smegelsky

The Earth observation market is at the very centre of the evolution of the space sector in a new era. Rapid adoption of innovations, lower cost, and easier access to space offer new opportunities for the private sector to enter a market once dominated by institutional and national programmes. Small satellites are the pivot around which this evolution is taking place. The European Space Agency has fostered a number of technologies and systems to enable European industry to compete in this market. The most recent of these initiatives is MicroMHiDe, a technology development to achieve sub-meter resolution from a small satellite, a task that was considered out of reach for a small satellite. MicroMHiDe includes a Korsch telescope and a new large format CCD on CMOS detector. MicroMHiDe uses a very aggressive implementation of the Korsch classical optical design resulting in a high performance and extremely compact system. It uses innovations in manufacturing processes and a novel implementation of the design. A new metrology system to perform on-board real time image quality correction is currently being studied. Furthermore, the new multi-spectral detector implementing CCD on CMOS allows time delay integration boosting image sharpness. This solution, developed within a GSTP activity with a consortium led by Belgian firms Aerospacelab, AMOS and IMEC, has been conceived with a design-to-cost approach that minimises the recurrent costs in view

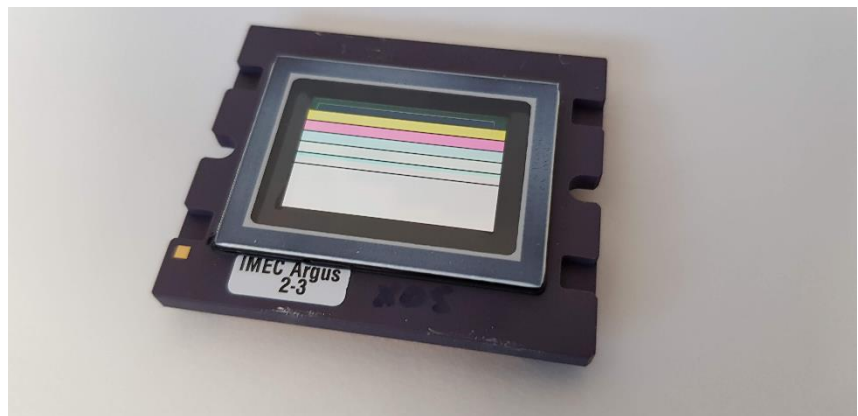
of deploying a constellation of small satellites. Fast revisit time, sub-meter resolution, high image quality, affordable cost and debris mitigation are requirements that seemed altogether unachievable but are now within reach with the implementation of the technologies developed with MicroMHiDe.

MicroMHiDe Technical Characteristics

Ground Sampling Distance	0.7 m
Swath	2.5 km
System SNR	> 100
Spectral Range	400 – 1000 nm
Spectral bands	1 PAN + 6 Multispectral
Mass	20 kg
Volume (Diameter – Length)	300 mm - 350mm



*Image credit: AMOS -
MicroMHiDe Telescope*



*Image credit: IMEC - Argus CCD on CMOS
Panchromatic and Multispectral Detector*

The Analog-1 test project, concluded with a two-hour space-to-ground test on 25 November 2019, had multiple technical goals. High among them was to assess the use of ‘force-feedback’ controls, giving astronauts a sense of touch in space, to evaluate if this technology would enable high-precision robotic control in weightless conditions. The hardware and software used by Luca Parmitano aboard the ISS, and the rover he controlled on Earth, were developed by ESA’s Human Robot Interaction Laboratory, in TEC, with the support of the DLR Institute for Robotics and Mechatronics – including the integration of control software to customise the system’s force-feedback and haptic feedback for the inescapable time delay experienced within the system.

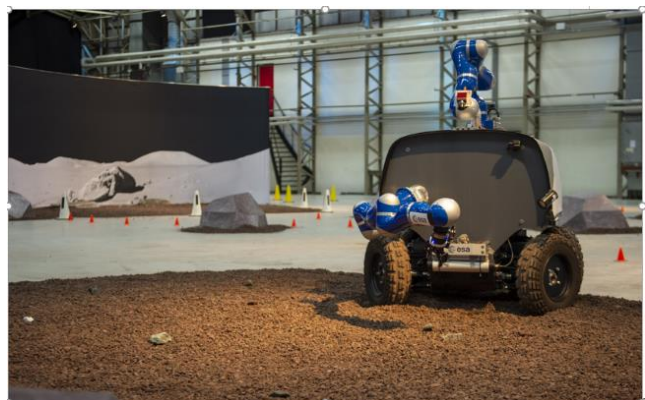


Image credit: ESA/G. Porter – Analog-1

In 2019, the ESTEC Test Centre tested many components and sub-systems for the Juice and MTG programs. The METOP SG STM qualification test campaign was also successfully executed together with some delta qualification for the Vega evolution program. Resources were carefully used so as to maintain all the facilities operational and improve them where possible. Maintaining the reliable status of the test centre facilities is a priority and is being achieved through the year by in-house engineering and industry support.

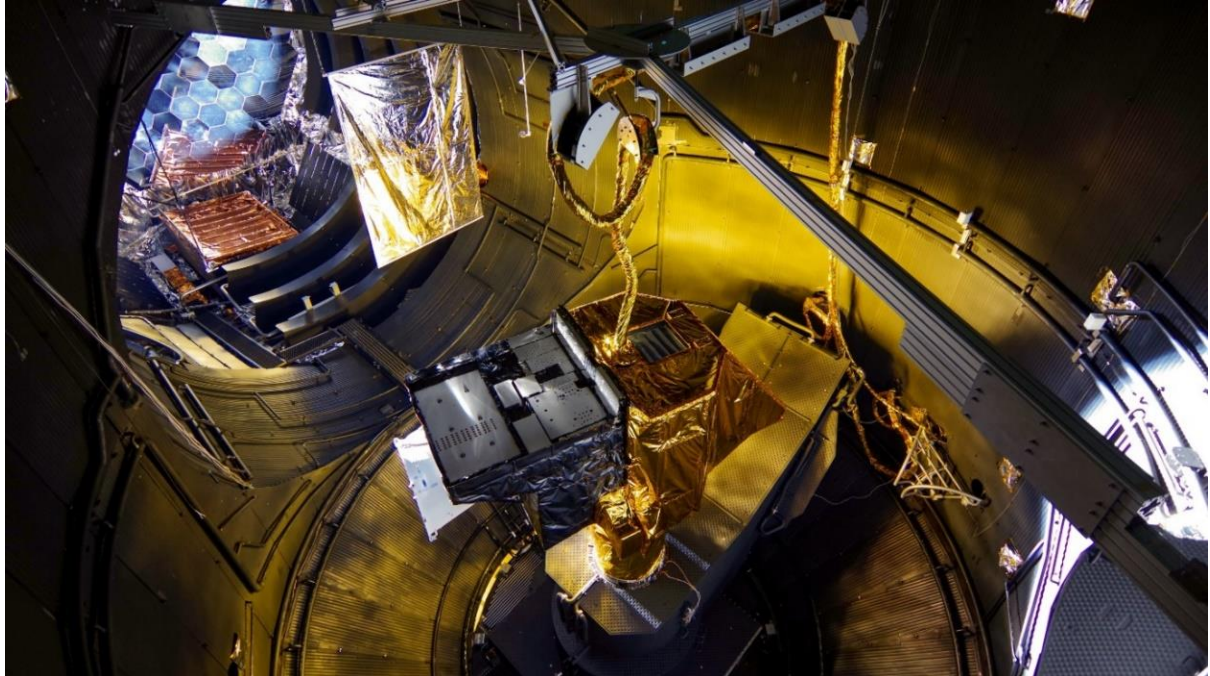


Image credit: ESA, MTG-IRS

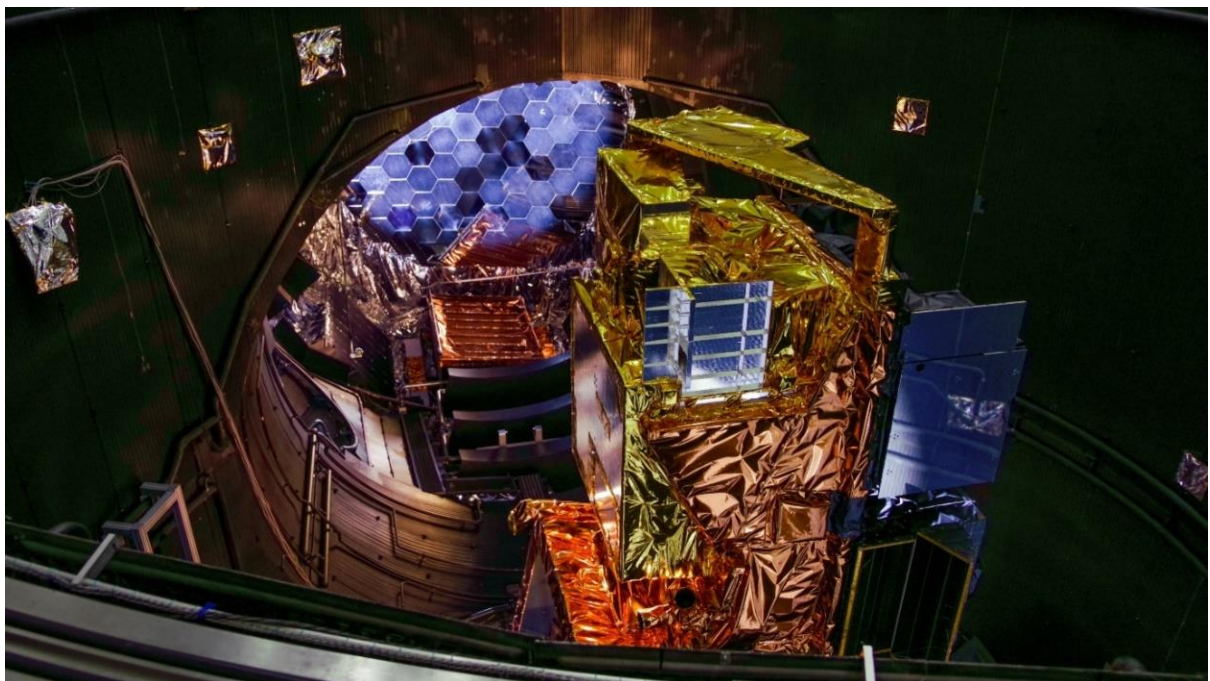


Image credit: ESA, METOP-SG LSS



Image credit: ESA- Vega-C fairing's trial by sound

ESA's Advanced Manufacturing Cross Cutting Initiative has the main objective of identifying and spinning in the space business disruptive materials and manufacturing processes readily available in the current European industrial landscape. With European leadership on the strategic manufacturing capabilities, the opportunity is there to stimulate the space industry supply chain, improving cost, schedule and sustainability, maximising the performances of the final space products, and even enabling new mission concepts. The first GSTP compendium for Advanced Manufacturing was issued in 2015, and additional activities were proposed through complementary ESA funding schemes. To date, over 80 technology development activities with an overall volume of €53m were committed, with contributions from all ESA Member States.

The recent introduction of Industry 4.0 has identified new opportunities, which are largely derived from the automotive and aeronautical industrial sectors. This includes the possibility for the European space industry to adopt manufacturing digitalisation and automation (robot-assisted production), Smart factory production-line simulation and machine learning, hazard monitoring and predictive maintenance, big data-driven production, verification and quality control. These technologies are identified in the GSTP compendium issued in 2019.



759-gram telescope composed of one casing and 2 mirrors made of AlSi10Mg additively Manufactured - compared with the 2.8 kg conventionally manufactured benchmark, It was designed, manufactured and tested within 2.5 years and fulfils all the relevant requirements for the intended application (credit TNO).

The TEC lead project SCouP demonstrates how a small team with support from ESA's Research & Development Technology programmes can achieve a real-life 'scoop' – creating a product that's a European first using shape memory alloy technology in an innovative manner. In every satellite propulsion subsystem, there are critical welds, which are very costly to perform – as well as to repair if they are not perfectly executed. Repair work can also be tricky, due to inaccessibility or for other reasons – necessitating specialised orbital welding and X-ray inspections/3D-scans. SCouP has produced a technology that makes it possible to quickly couple two stainless steel pipes with a level of tightness that is comparable to a weld. The success of this development has already raised the interest of all of the European primes and NewSpace actors with possible use in cubesat-like applications. Moreover, technology transfer opportunities have opened to spin off the technology to non-space applications – there is interest for non-space applications such as the hydraulic lines of helicopters as well as medical applications.



Image credit: First SCouP (Shape Memory Coupling for connecting Pipes) production batch – Credit SCouP s.r.l. (Belgium)

The TEC Mechanisms Section has developed and launched a gear analysis facility using state-of-the-art software primarily developed for the automotive industry. This facility is now fully operational and has already supported the JUICE SADM and Euclid VIS instrument, Read Shutter UNIT (RSU) solving major issues, determining root causes of major gear anomalies and supporting gear redesign. In addition, in cooperation with the European Space Tribology Laboratory (ESTL), work is ongoing in order to further correlate the analytical results of the analysis facility with physical test results. The outcome of this correlation will aid future missions in providing much more accurate predictions of gear life time.

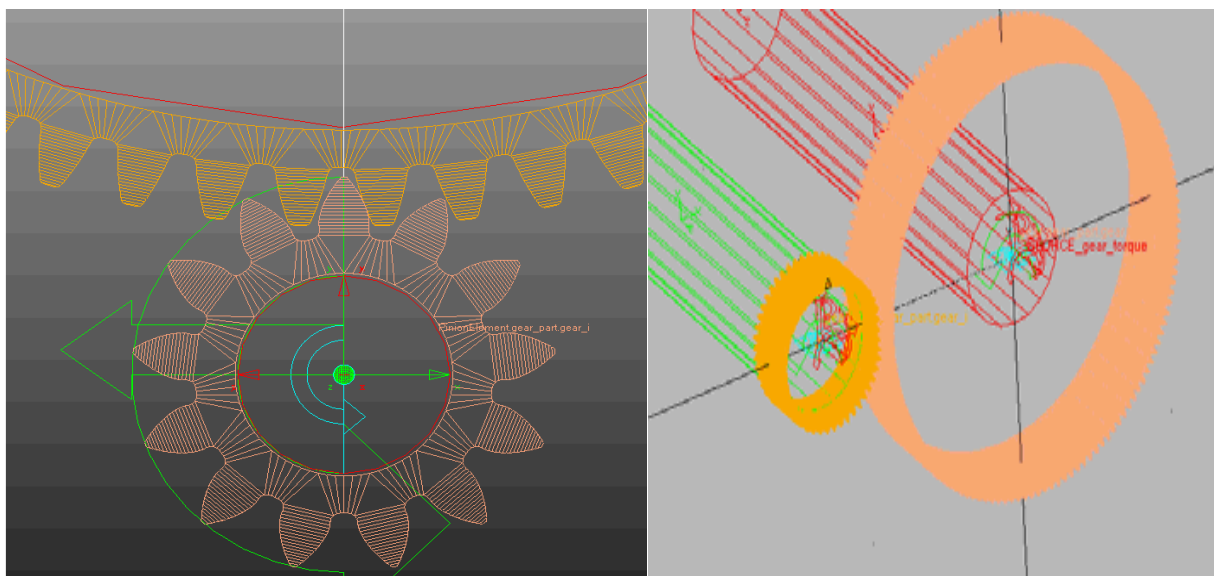


Image Credit: ESA/Mr. Sandro Patti - Juice SADM Adams GearAT model analysis

To fulfil the stringent Clean Space requirements of passivation at end of life, new European pyrotechnics initiators have been qualified in 2019 for a lifetime of over 20 years in space. Typical pyrotechnics initiators are limited to 8 years. These new initiators will be used on the JUICE mission to actuate a pyro-valve after 13 years in space. These initiators fully comply with the REACH regulations.



Image Credit: Dassault (FR) - ZPP/RDX squib equipped with safety cap

After major anomalies occurred during on-ground testing of the Vega Zefiro 40 at the end of 2017, a new development, started at the beginning of 2018, concluded at the end of 2019 with the qualification of a new detonator for Vega and Ariane. The modifications consisted of a change in the shape of the detonator cap, a change in the explosives separator sheet and in the initiating powder, the latter to comply with Clean Space and REACH requirements. A partially modified detonator (with changed cap shape) was successfully introduced already from the Vega VV13 to Ariane5 VA253 flights. The first flight with the fully modified detonator (including the other two modifications) will be Vega VV16 scheduled for September 2020.

Resources Management, corporate activities and external relations

Human Resources, Facility Management and Informatics

Human Resources

Staff Engagement Survey

ESA places a high value on staff engagement and its beneficial impact on organisational performance. Accordingly, a staff engagement survey was conducted in 2016 and repeated in 2018 as a joint initiative by the Administration and Staff Association.

The surveys are a strategic resource for the Agency in its efforts to shape a satisfying, supportive and inspiring working environment, and to ensure ESA's future success by identifying its strengths as well as areas in need of improvement, on the basis of regular feedback from staff.

The results of the two surveys have spawned various initiatives to improve working life at ESA that have been implemented in recent years, taking as just a few examples adjustment of the teleworking policy to allow for greater flexibility, the "ESA-inside" initiative to ensure a continuous dialogue between management and staff, optimised recruitment, payment and procurement processes. All directorates together with the Administration are continuously following up on ongoing activities in response to the survey results.

Learning & Talent Development (L&TD)

The Administration offers a training catalogue with over 200 internal courses. Some of these courses are very popular and sought after. In addition, external courses are requested by staff and managers.

To manage all of these courses more efficiently, an online learning administration module (part of the hr-space SuccessFactors suite) was implemented in March 2018 and has since been adopted by all Agency staff and contractors without any major hitches.

Entry-Level Programmes

ESA offers a variety of entry-level programmes — such as Young Graduate Traineeships (YGT), Research Fellowships (RF), National Traineeships (NT), and Internships. Each programme has its own purpose and advantage from which both ESA and the trainees benefit.

In 2019, ESA hosted 263 YGTs, 51 NTs, and 79 RFs. The YGT 2019 recruitment campaign was particularly successful. Not only has the number of applications increased substantially to reach 7820, but the Agency attracted an increased number of women and candidates from under-represented states with 46% of all hired YGTs being women and 85% of all hired YGTs from under-represented states.

Performance Reform

ESA has been undertaking a major reform of performance management in the Agency. The reform is based on two drivers: 1) a commitment to Member States to better control the wage bill and harmonise the grading structure, and 2) a dissatisfaction with the level of transparency and fairness of current practices, expressed by staff in the staff engagement surveys of 2016 and 2018.

Following extensive negotiations between the Administration and Staff Association, the Staff Regulations and Rules were revised and new Instructions established. All the relevant ESA bodies approved the entire legal framework in 2019.

Due to the very substantial cultural change this reform requires, a phased implementation approach is planned with a trial year in 2020, an effective date of 1 June 2021 and the first merit recognition exercise under the new system to be conducted in February to May 2022.

Working arrangements

In 2019, on top of the Performance Reform Policy, HR also introduced Mediation and Harassment policies. In addition, ESA engaged conducted a modernisation of its Internal Justice System. The amendments were finalised in 2019 and are expected to enter into force in 2021 following AFC and Council approval.

Finally, increases were adopted in the contribution rates to both Pension Schemes. For the Coordinated Pension Scheme, the change was linked to a revision of the adjustment method and the entitlement conditions to the education allowance. In 2019, the number of pensioners increased by 71, to a total of 1595 at year's end.

Diversity & Inclusiveness (D&I)

A word from ESA's Chief Diversity Officer: In 2019, in the framework of the Diversity and Inclusiveness initiatives, very positive results have been achieved in attracting and retaining a gender-balanced and geographically diverse workforce. The Agency has also reinforced its engagement in reaching out to young generations, and actions have been taken to offer opportunities to young people with physical disabilities and to foster an inclusive working environment. ESA's commitment to diversity and inclusiveness has received increasing international recognition, culminating in the Agency winning the IAF Excellence in "3G" Diversity Award for its contribution to promoting the values of diversity in the space sector.

Estates and Facilities Management

ESA's Estates and Facilities Management (EFM) Department is responsible for the management of much of the infrastructure and services that underpin the daily life of the Agency. From securing the perimeters of its sites and centres to maintaining its vital technical systems, its span of activity is very wide. 2019 was a typically busy year with significant progress on a number of high-profile construction activities and the renewal of a number of key contracts.

The project to renovate ESA's Headquarters passed a number of significant milestones. The turnkey construction contract was signed in May based on a vision of creating a modern home for delegations, the Director General and key staff that reflects the new digital world of 2022/23 when it is due to be completed. The project is complex both technically and because of its context within the busy centre of Paris. The ESA Council was kept fully informed of progress throughout the year.

The integrated facilities management contract for ESOC was re-tendered and a new commercial partner appointed. Within the typically tight financial constraints, the ESOC team was able to build in higher standards of output that will benefit both ESOC's technical infrastructure and the customers of the many different FM services within the contract.

In ESTEC, the 'soft-services' FM contract went through a similar procurement process and the outcome was again a change of incumbent and the introduction of testing KPIs and outputs.

Underpinning all of the activities of the EFM Department was a determination to improve the environmental standards of all managed processes. A dedicated team oversaw the installation of a wide range of low-carbon-emitting systems both in new construction and as part of refurbishment programmes. Waste streams were optimised and formal certification broadly introduced. All of this designed to ensure that ESA's internal operations match the Agency's aspirations for stringent environmental performance in all aspects of its activities.

Information Technology

Significant efforts were made in 2019 to upgrade, improve and provide extensive digital options for ESA's daily business placing it in good stead as it faces challenges both present and future.

ISO 27001 certification

The ESA IT Department has been successfully accredited with ISO 27001 certification. This industry standard of security certification is awarded to organisations, or departments within organisations, that demonstrate both secure infrastructure and proactive digital security practices. This was the culmination of a longer effort which changed the ways that the ESA IT Department proactively fights cyber-crime. This certification now improves data-sharing with ESA's external partners.

ESA official documents

Developed by the IT Department and the Director General's Cabinet, esa blue-docs has now replaced eCPB. All documents required for Member States to take decisions on ESA activities are digitally accessible to delegates and Agency staff. This new paperless system provides Member State stakeholders with a secure, digital solution that is considerate towards the needs of the stakeholders and the environment.

Digital provisioning tool

The IT Department has upgraded the digital provisioning tool 'esanow', which allows staff and contractors to request IT resources. Approvals are digital and automatic. Changes made in 2019 also ensured that the supply chain was integrated into the mechanics of the software, further automating this process.

Workforce Management

The ESA IT Department collaborated with the ESA HR Department to create new tools and services for the management of ESA assets and manpower. A new timesheet tool for contractors, and reporting modules in the training solution within 'hr-space' round out the current digital suite. The Programme, Project, Strategy tool has been prepared to go live – providing dedicated reporting for ESA Member States.

Economic Printing Services

A new corporate printing service has been activated at all ESA sites to reduce the use of personal printing devices and the number of printed pages. This service reduces waste from uncollected printing and redundant personal devices producing a significant cost saving, and lessening our environmental impact.

Preparations for esa365

The ESA IT Department has prepared the Agency for migration to esa365. This required a sustained effort in 2019. This entails a careful migration of existing ESA data to the Microsoft Office 365 offering. It is being done in a secure and user-friendly manner and will begin rolling out in 2020. This has only been possible with a number of other improvements to areas such as security and access management also delivered in 2019.

Procurement and Industrial Policy

Procurement

Next to contract management of many large complex projects (MTG, METOP, Biomass, Earthcare, FLEX, Sentinels 1-6), 2019 was largely devoted to elaborating a procurement strategy for the High Priority Copernicus Missions (HPCMs) and obtaining Member States' and European Commission agreement on that procurement approach in anticipation of Space19+ decisions. The six Invitations To Tender with many innovative procurement policies and contractual features were been prepared and issued.

In addition, procurement activities supported the preparation of the Transition / 2nd Generation of Galileo with dialogue meetings served to refine the requirements against which industry prepared its final proposals for the Galileo Transition Satellites (now called Galileo 2nd Generation Satellites Batch 1). The Galileo transition payload test bed contracts were kicked off contractually in Summer 2019.

For the current generation of Galileo, the contract for the new dual launch dispenser development and qualification for the Ariane-62 launcher were kicked off with Ariane Group in August 2019 while a complex ITT was also prepared on Satellite Insurance for Batch 3 Satellites.

Numerous ITTs/RFQs and related contracts for NAVISP elements were also processed, as well as some H2020 Navigation activities. For a new EU-funded activity (under Horizon 2020), the supply of Carriers

(Micro-, Mini-, CubeSats) for the aggregation of IOD/IOV Experiments was the subject of two separate ITTs and their evaluations (in parallel to preparing a number of Memorandums of Understanding with Experiment Providers).

The Phases C/D/E/F contract for the PROSPECT Development for Lunar Exploration was signed with LEONARDO SpA in December 2019.

In 2019, the Industrial Policy Committee (IPC) approved the contract proposal for the procurement of the third European Service Module (ESM3), which was duly carried out and the negotiations were also started in the course of the year. In addition, the IPC authorised the request of an industrial offer for the Batch Procurement of 6 European Service Modules (ESMs 4 to 9) and the request was released to industry. Additionally, the ITTs were issued for Lunar Gateway and the Mars Sample Return missions following IPC approval of the procurement proposals.

The down-selection for the Implementation Phase (B2/C/D/E1) of the Solar wind Magnetosphere Ionosphere Link Explorer (SMILE) Payload Module (PLM) was successfully completed together with the Rider with Airbus Defence and Space S.A.U (E) and signed in July 2019. Following IPC approval of the procurement proposal and issuance of the relevant ITTs, Phase parallel contracts were placed for each of the three M5 candidate missions, namely THESEUS, EnVision, and SPICA.

In line with the decision taken at the Intermediate Ministerial Meeting in autumn 2018, IPL developed a new “grant-like” procurement tool to provide ESA Initial Support to Innovation (EISI). The relevant legal framework resulting from a close collaboration between the Legal and Procurement Departments, was approved by the ESA Council in December 2019 upon recommendation of the IPC and AFC. A pilot phase for several programmes has been agreed to be initiated for a period of two years.

Finally, the improvements and enhancements to the esa-star electronic-tendering system continued in the course of 2019 leading to two major system releases.

During 2019, two important additions to esa-star functionalities were released, namely the management of tender actions, including CCB and CRB processes, functional to the placing of Contract Changes (CCNs, Riders, etc.) as well as the management of (Restricted Competitive) Tender Actions under existing Frame Contracts. The two functionalities were opened for pilot testing with the aim of proceeding with a general opening in 2020.

In 2019, the Analysis and Design phase for “esa-star / Publication”, the new system which will replace EMITS as of Q1/2021, was completed.

Industrial Policy and Audit

The Industrial Policy and Audit Department contributed substantially to the preparation of Space19+ and to its success. Ahead of the Ministerial Council, the Department elaborated together with the Programme Directorates the specific industrial policy texts to be included in Programme Declarations and Implementing Rules, preparing also elements of a Resolution on the evolution of ESA's industrial policy. It must be emphasised that in 2019, most of the new ESA programmes presented for decision at Space19+ were supported by a socio-economic impact assessment led and coordinated by the Department, with a total of ten independent socio-economic studies by leading European consultancies. The studies covered all four pillars of ESA activities, representing a significant improvement and unprecedented effort compared with the previous ministerial meeting. The Space Economy Website (space-economy.esa.int) also provided support to the Member States benefitting

from the Industrial Incentive Scheme for the preparation of Space19+. As a result, those Member States have either maintained or significantly increased their contribution to a level commensurate with the size of the country. The programmatic targets of the New Member States Incentive Scheme Calls have been aligned with their (intended) subscriptions to ESA, thereby promoting industrial competitiveness in the optional programmes in which they participate and helping ensure proper geo-return.

Space19+ preparations were also supported by the ESA Industrial Relations Office and SME Office. Highly successful events such as the Industry Space Days and the 2019 High Level Forum with industry gathered hundreds of industrial participants and allowed fruitful exchanges to take place among firms (of all sizes, geographical origin and from all disciplines), European institutions and Member States. The SME Office initiated the development of two new online platforms — the “Learning Hub” as an online environment to provide access to training courses for industry, and “esa-match” as a matchmaking tool for industry to be operational in 2021. The SME Office also continued its cooperation with the European Investment Bank concerning the facilitation of access to financing for SMEs.

2019 was the last year of geo return statistics for the Formal Review of geo-return during the 2015–2019 period. During this period, the Executive deployed reinforced proactive industrial policy and geographical return management mechanisms in order to ensure sustainable attainment of the quantitative and qualitative objectives and to avoid resorting to geo-return discontinuation in the future. These efforts proved extremely effective with unprecedented results attained for the first Formal Review of a new statistical period: results better than the overall guaranteed minimum criteria were secured for Member States by the end of 2019. This has also benefitted from the new ESA Return Operating System (EROS), which provides full transparency and performance reporting to ESA Member States, programmes and industry.

Finally, in 2019, the industrial auditors managed to achieve substantial savings across ESA Programme Directorates. This has been possible thanks to major complex rates audits achieving significantly lower results than those proposed by Industry/National Audit Authorities. Furthermore, concrete changes have been introduced to the ESA regulatory framework (ESA GCC) to bring transparency and equal treatment across industry in the different Member States with Council approval in June 2019.

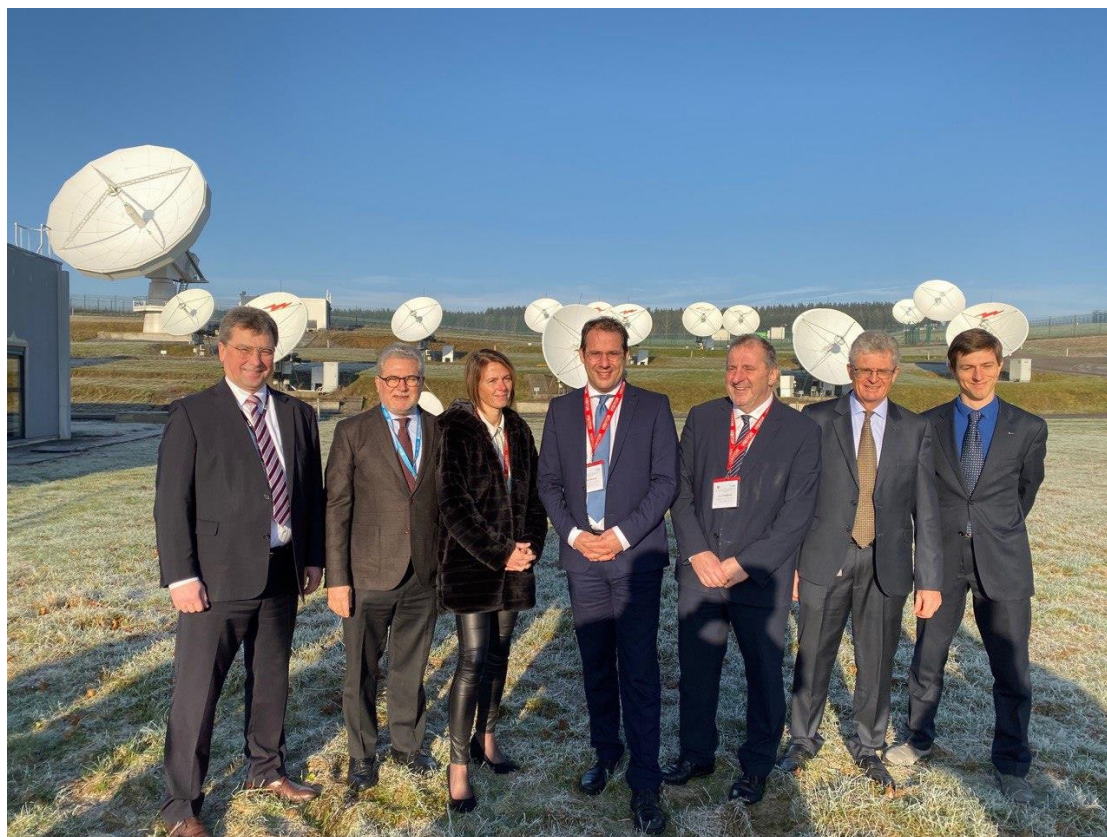
Compliance, Quality and Ethics Office

The Compliance, Quality and Ethics Office is responsible for defining, implementing and monitoring an ESA-wide compliance programme, based on compliance risk assessment. In 2019, compliance activities were mainly focussed on implementing the ESA Personal Data Protection Policy and further developing Agency-wide awareness of ESA Policy on the Prevention, Detection and Investigation of Fraud. The Office is also responsible for ensuring the implementation and maintenance of the Quality Management System of the Administration Area. In 2019, emphasis was placed on further embedding the principles of quality management into daily practices and digitalising process-related information.

ESEC

Building on the successful operation since 2016 of the first ESA cyber range at ESEC, a proposal to further enhance ESA's cyber resilience and to host ESA's Cyber Centre of Excellence and Cyber Security Operational Centre at ESEC was strongly supported by Member States at Space19+ and particularly by the host country, Belgium, which agreed to make a significant contribution to this effort, thereby supporting a decision fully in line with the Centre's security mandate in the overall context of ESA. Inaugurating a new state-of-the-art infrastructure at ESEC just after Space19+, the Belgian Minister

responsible for space underlined the country's commitment to supporting European space activities from Belgium.



Mr Ditter – E. Morel de Westgaver – Anne Laffut – Minister David Clarinval – Minister Willy Borsus – E. Tilmans – R. Wittmann



Visit to the ESEC Cyber Training Centre in presence of ESA Director E. Morel, Astronaut T. Pesquet, Governor of the Belgian Province of Luxembourg, O.Schmitz, Minister Bellot, Minister Borsus, Mayor of Libin, A. Laffut, Head of the Belgian Delegation F. Monteny, Presenter: D. Wiemer (RHEA)

Finance, Planning and Control

Financial Accounting & Operations

ESA successfully completed the closure of the 2019 accounts according to the new accelerated timetable as specified in the revised Financial Regulations, following the reform of the audit system decided by Council in 2016. For the eighth consecutive year, the Agency received a "clear" audit opinion, a major achievement that provides maximum reassurance on the quality of its financial management, administration and reporting, and reinforces the foundations of trust and confidence.

Efforts in 2019 focused on the consolidation of the many improvements implemented in previous years. Following the Audit Commission's recommendation to further strengthen the Agency's internal control system in order to meet audit readiness objectives and the revised timetable, the Agency successfully completed a second interim closure. The Executive also further automated the Financial Statements in order to accelerate their production and the layout of the Financial Statements was further streamlined.

In addition, following a recommendation from the External Auditors, the Executive successfully implemented an accounting risk assessment review of significant purchase contracts.

The accurate management and accounting of the Agency's payments represented a major (ongoing) undertaking, consisting of some 70 000 transactions related to payments and mission settlements.

Finally, it is worth noting that significant progress was made in 2019 in the further digitalisation of financial operations, placing ESA in a position to be able to handle the full-scale processing of payments fully electronically via teleworking, if required.

Corporate Planning, Budgeting and Controlling

Corporate planning and control played a key role in 2019 in consolidating and assessing multi-year financial plans of all programmes and activities, leading to a successful recommendation of cost plans in all Programme Boards and a unanimous adoption of the 2020 budgets in Council. Budgets were prepared in discussion with contributors, taking into account their funding capacity in view of ensuring a balance between cash outflows and available resources.

The planning and controlling function made a significant contribution to the preparation of the Space19+ ministerial meeting in Seville, with the preparation and establishment of sustainable plans, support to the definition of new programmes and activities, preparation of the Director General's proposals on the mandatory activities and strategic financial advice to the Director General and Executive Board. Corporate planning and control also provided substantial support and advice to States regarding possible subscriptions, simulations and affordability requirements.

In 2019, it also played an integral part in the preparation of the Digital Subscription Tool for Space19+. The tool was created, built and utilised within several months, thereby enabling States to subscribe to newly proposed and ongoing ESA programmes and activities.

Continued progress was achieved in the development of an Agency-wide tool for planning and control with the initial internal rollout of the three 'Cluster 1' Directorates; rollout of the remaining Directorates is now being planned.

Cost

ESA's total cost in 2019 amounted to €5103m, of which €3840m was related to ESA programmes and activities and €1263m to Third Party activities.

Mandatory Activities represented 16% (€825m), 56% was accounted for by Optional Programmes (€2881m), 2% by activities associated to the General Budget (€113m), 25% by programmes financed by Third Parties (€1263m) and 0.4% by others (€21m).

The core elements of Mandatory Activities are the Scientific Programme and Basic Activities. Basic Activities represented 5% of ESA total cost in 2019 (€250m) and included, among other things, the Discovery, Preparation and Technology Development Programme, Core Technical Expertise, Common Infrastructure, Technology Research Programme, Earthnet and Long-Term Data Preservation. In 2019, the Scientific Programme cost amounted to €575m (11% of ESA total cost). Main activities were the development of JUICE, Solar Orbiter, EUCLID, PLATO as well as the preparation of future missions (Voyage 2050). CHEOPS was successfully launched on a Soyuz-Fregat from Kourou in December 2019 and BepiColombo continued its transit towards Mercury. In addition, the Scientific Programme funded the operations of 11 satellites in orbit, among which GAIA reached the all-time record of scientific publications in a year. The Prodex programme continued throughout the year with institutes and industries participating in over 250 ESA activities in the domains of Space Science, Earth Observation, Microgravity and Exploration (scientific instrumentation and exploitation).

The execution of programmes including the study, development, launch and operations of satellites and spacecraft as well as the development, exploitation and maintenance of launchers and other space systems is implemented via the Agency's Optional Programmes, in which Member States participate through a voluntary subscription.

In 2019, €1044m, 20% of ESA total cost, were accrued on Space Transportation programmes (excluding the Guiana Space Centre, CSG). Ariane exploitation activities supported four successful launches of Ariane 5 during the year. The development of the Ariane 6 Launcher System reached its final phase. The activities on the Ariane 6 Launch Base are progressing towards Launch Base Technical Review. On the Vega C development side, the Final Project Key Point took place in December, providing a final definition of both the Vega-C configuration and high-level schedule. For Space Rider, the main Critical Design Reviews (CDRs) at subsystem level were completed in preparation of the system level CDR. In addition, Future Launchers Preparatory programmes continued to advance on various technologies and integrated Demonstrators according to their objectives.

In Earth Observation, €727m was accrued in 2019, 14% of ESA total cost, and covered the continued development of the Sentinels, as well as the meteorology satellites with activities for Meteosat Third Generation and MetOp Second Generation. For Biomass, development of the flight hardware got under way and the launcher was selected. Joint work with JAXA on EarthCARE continued, and the contract for the launcher was placed. Negotiations were finalised for the ALTIUS Phase B2/C/D procurement. FORUM was selected as the ninth Earth Explorer.

Cost for Human Spaceflight, Microgravity and Exploration reached €483m, 9% of ESA total cost. The Orion-ESM costs reflect the schedule robustness measures agreed with NASA in the course of 2019 leading to achievements such as the ESM1 PTR integrated system test at KSC and the delivery of water tanks for the ESM2 among others. Similarly to ESM1 and 2, the Orion-ESM 3 cost includes the ESM3 structure successful CDR, MRR and ARR. Furthermore, cost includes partial payments agreed in the frame of the December 2019 agreement for ESM 3 and ESM 4-6 batch following Space19+ decisions

on the continuation of the programme. In the area of ExoMars, cost accrued covered the preparation of the activities for the 2020 launch such as the Rover Module FM environmental test campaign, the SpaceCraft Composite (SCC) thermal vacuum test and the spacecraft Assembly Integration and Test (AIT).

In Telecommunications and Integrated Applications, representing €400m, 8% of ESA total cost, the main objective remains European competitiveness on the worldwide market and this was implemented through various programme elements: Core Competitiveness, several Partnership Projects, Business Applications and Space Solutions as well as the crucial Future Preparation. As one programmatic highlight, the EDRS-C satellite was successfully launched in August 2019. Space19+ was a success for the ARTES programme, enabling continued future support to European industry in the Satcom market.

In 2019, the cost for Technology Support represented 2% of ESA total cost (€125m) €22m of which was for Proba-3 development. The remaining costs were applied to R&D activities.

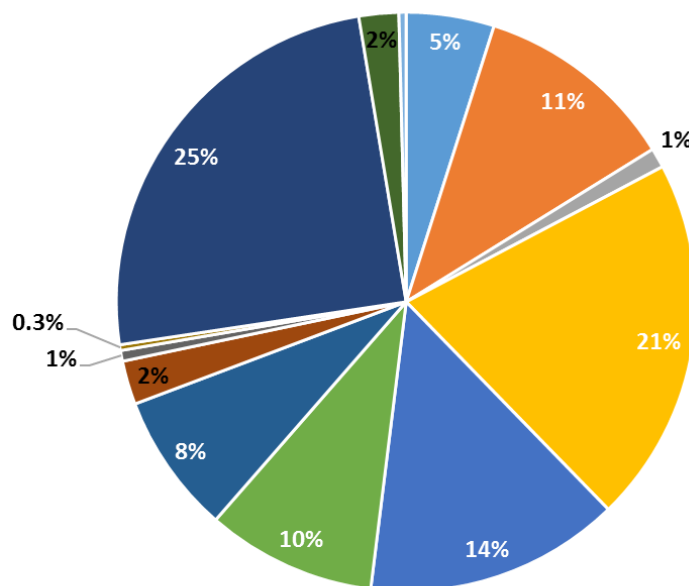
Cost for the Space Situational Awareness Programme amounted to €31m in 2019 (1% of ESA total cost). Two hosted space weather payloads became operational and performed excellently, one of the planetary defence robotic telescopes became operational and the Fly-Eye telescope camera prototype successfully passed the Final Acceptance Test.

The Navigation programmes represented 0.3% of ESA's total cost in 2019 (€16m). The NAVISP Programme has now incubated more than 100 activities. At Space19+, Ministers gave a clear vote of confidence in the NAVISP programme by subscribing to the phase 2 with a significant enhanced yearly level of activities, providing the means to increase the volume of activities.

For Third-Party activities and European Cooperating States Agreement (ECSA), there were €606m related to Navigation activities funded by the European Union through the Galileo and EGNOS Delegation Agreements and the EGNOS GSA Working Arrangement. Following the Galileo service disruption in July 2019, work is in progress to recover nominal redundancy. Redefinition of the Galileo FOC (Full Operational Capability) schedule is being undertaken to take due account of impacts of the incident as well as the results of the Inquiry Boards and the Lessons Learned. The Galileo programme is further proceeding with the development of 12 batch#3 satellites. With regard to the Galileo Transition Satellites (GTS), procurement is at the stage of competitive dialogue. System Requirement Review phase 2 (SRR2) was due to be kicked off at the end of January 2020. G2G preparatory R&D activities under H2020 are continuing as planned, and on EGNOS first generation, the next major update (V2.4.2.B) Phase C/D contract has been signed. EGNOS next generation development (V3) is still adhering to the committed schedule thanks to risk mitigation actions. In the Earth Observation area, Third-Party activities amounted to €572m, mainly related to cooperation with the EU on Copernicus (€423m). The remaining cost is linked to cooperation with EUMETSAT on MetOp and Meteosat Third Generation. SeoSat Launcher procured, with the launch planned in 2020.

Cost (€million)

- Basic Activities €250 million
- Scientific Programme €575 million
- Prodex €56 million
- Space Transportation €1,044 million
- Earth Observation €727 million
- Human Spaceflight, Microgravity & Exploration €483 million
- Telecommunications & Integrated Applications €400 million
- Technology Support €125 million
- Space Situational Awareness €31 million
- Navigation €16 million
- Third Party & ECSA €1,263 million
- Associated to General Budget €113 million
- Miscellaneous €21 million



Income

ESA Mandatory Activities (including the General Budget and Scientific Programme) and activities associated with the General Budget are financed by Member and Participating State contributions. The ESA Convention stipulates that the Agency's mandatory activities are financed using a contribution scale based on the national income of the Member States over the previous three years.

ESA Optional Programmes are financed by Member and Participating State contributions, and for some programmes, also by other entities, including the European Union and EUMETSAT. Participating States declare a voluntary subscription on a multi-annual basis.

Third Party Programmes, managed by ESA, are fully financed by third parties. In 2019, they represented 27% of total ESA income.

	2019	
	Mandatory Activities (%)	Optional Programmes (%)
Austria	2.17%	1.51%
Belgium	2.72%	5.15%
Czech Republic	0.95%	0.80%
Denmark	1.82%	0.51%
Estonia	0.13%	0.05%
Finland	1.34%	0.26%
France	14.33%	31.95%
Germany	20.70%	23.21%
Greece	1.15%	0.03%
Hungary	0.63%	0.10%
Ireland	1.09%	0.35%
Italy	10.52%	12.65%
Luxembourg	0.20%	0.87%
Netherlands	4.64%	1.20%
Norway	2.72%	1.39%
Poland	2.87%	0.59%
Portugal	1.12%	0.27%
Romania	0.99%	1.15%
Spain	7.12%	4.40%
Sweden	2.71%	2.04%
Switzerland	4.07%	3.82%
United Kingdom	16.01%	7.36%
Total MS	100%	99.69%
Canada	4.26%	0.25%
Total Cooperating	4.26%	0.25%
Slovenia	0.16%	0.06%
Total Associate MS	0.16%	0.06%
ESA	104.42%	100.00%

Budgeted contributions from Member and Participating States in 2019 to mandatory activities and optional programmes

Workforce Management

2019 was the third year since the flexible approach to the staff complement was introduced. The flexible utilisation of the staff ceiling has allowed the Agency to advance first recruitments to address the departures wave of the coming years, to manage workload peaks and to absorb new activities from 2016 Ministerial Council as well as the sudden increase of workforce needs related to the Galileo Transition Satellites. These achievements have led the Agency to propose an increased flexible approach following the 2019 Ministerial Council at which the flexibility range was increased to 7% in 2022. The contractor workforce, both loan employment and service, has steadily grown and the Agency has reviewed all the main contracts to reinforce compliance with national labour legislation.

Member States	Hors Classe	A	L	B	C	Total Staff
Austria	1	34	1	2		38
Belgium	1	102		4		107
Czech Republic		6				6
Denmark		10				10
Estonia		2				2
Finland		18		4		22
France	2	435	9	67	4	517
Germany	3	412	4	42		461
Greece		25		2		27
Hungary		4				4
Ireland		16		5		21
Italy	1	381		18		400
Luxembourg		5				5
The Netherlands	1	64		10		75
Norway		24		2		26
Poland		21		1		22
Portugal		29		2		31
Romania		16		1		17
Slovenia		1		1		2
Spain		197		3		200
Sweden		41		2		43
Switzerland	1	26		1		28
United Kingdom	1	236	3	17		257
Total	11	2105	17	184	4	2321
Non-Member States						
Canada		20				20
Total		20				20
Grand Total	11	2125	17	184	4	2341

Legal Services

In 2019, the ESA Council took a large number of decisions, among which the adoption of the:

- Resolution on “Space as an Enabler”, adopted at the 9th Space Council (ESA/CE-M/CCLXXX/Res.1(Final));
- Resolution on the ESA Initial Support for Innovation (“EISI”) (ESA/C/CCLXXXVII/Res.1 (Final));
- Resolution on the Institutional Exploitation of ESA-Developed Launchers and supporting Competitiveness (ESA/C/CCLXXIX/Res.1 (Final));
- Resolution on the financing scheme of the Agency’s HQ Project (ESA/C/CCLXXXVII/Res.3 (Final));
- Resolution on the Agency’s Geographical Return Coefficient (ESA/C/CCLXXXV/Res.1 (Final)).

In addition, at the Council meeting at ministerial level held on 27–28 November, Council adopted the following:

- Resolution on Space: The 5 dimensions of Space 4.0 (ESA/C-M/CCLXXXVI/Res.1 (Final));
- Resolution on the Scientific Programme and the Basic Activities for Space19+ Level of Resources for 2020-2024 (ESA/C-M/CCLXXXVI/Res.2 (Final));
- Resolution on ESA Programmes: addressing the challenges ahead (ESA/C-M/CCLXXXVI/Res.3 (Final));
- Resolution on CSG (2020-2024) (ESA/CM/CCLXXXVI/Res.4 (Final)).

At the same Council meeting at ministerial level, the Council took note of a total of 15 Programme Declarations revised or drawn up by the respective Participating States over the year, as the legal basis for the Agency’s optional programmes.

In addition, 17 international agreements were concluded with governments, international organisations and other institutions in 2019, including a new landmark cooperation agreement between ESA and the Government of Canada (ESA/LEG/496), allowing to continue a decades-long successful cooperation.

ESA also declared its adherence to the Declaration of Interest for the Space Climate Observatory (ESA/C(2019)74) and acceded to the Memorandum of Understanding with European Defence Agency (EDA) Member States on Cooperation on Cyber Ranges (ESA/C(2019)53).

The ESA Legal Services Department not only accompanied the drafting, negotiation and decision-making of those instruments, but also prepared a wealth of additional legal documents including programme implementing rules, totalling over 100 decision items across Council and its subordinate bodies in order to build the basis for decisions taken at Space19+.

ESA Resolutions adopted by the Ministerial Council and ESA Regulations are available on the “Law at ESA” pages and the “Space19+ blog” of the website.

Relations with Member States

The Director General visited Member States regularly to attend major events as well as appointments with the Ministers in charge of space matters, delegations, and to address, whenever possible, parliaments, industry, the press or universities.

In the framework of the preparation of the November ESA Council at ministerial level (Space19+), the Director General held bilateral meetings at ESA Headquarters with all Directors and the 22 Member States. These meetings allowed open exchanges on the preparation of Space19+ and the main concerns or challenges on both sides.

A review of the establishments, offices, laboratories and centres or entities of various natures that form ESA's presence, its tangible "footprint" in the 22 Member States has been performed in order to have a better overall view and to be able to act should there be any need for reinforcement.

Space for Earth (www.esa.int/spaceforearth)

Space supporting the UN Sustainable Development Goals (SDG)

(<https://www.esa.int/spaceforearth/sdg>)

A Memorandum of Understanding was signed in May with the UN Office of Outer Space Affairs (UNOOSA). ESA activities in support of the Sustainable Development Goals (SDGs), as exemplified in the unique ESA catalogue (<https://sdg.esa.int>) will be showcased in the UNOOSA Space Solution Compendium under preparation. This ESA catalogue was presented and demonstrated on various occasions and was very well received with the recognition that such a tool was indeed needed and did not exist elsewhere.

Space and the Arctic

(https://www.esa.int/Enabling_Support/Preparing_for_the_Future/Space_for_Earth/Arctic)

The work of the ESA Arctic Task Force with the interested Member States that started at the end of 2016 culminated in the integration in the programmes proposed at Space19+ and in the subscription of a demonstrator for an Arctic Weather Satellite constellation. This constellation, useful both for Arctic activities and the improvement of global numerical weather forecasting, was supported by the meteorological offices of all ESA Nordic States and by EUMETSAT.

Blue Worlds

(https://www.esa.int/Enabling_Support/Preparing_for_the_Future/Space_for_Earth/Blue_worlds)

Following requests from several Member States, a study was performed to map the ESA programmes capable of supporting Member States' interests and priorities in tackling maritime regions' challenges and objectives. As interest from many Member States was reaffirmed, starting with Portugal and Greece, who were the promoters of two initiatives, an ESA Blue Worlds Task Force has been created, for all interested Member States (19), tasked with reviewing existing programmes and user requirements, analysing gaps and proposing concrete actions before the next Council at ministerial level. In addition to the Atlantic and the Mediterranean Sea, it also covers the Baltic, the Black Sea, the North Sea and the Arctic Ocean.

ESA Lab@ initiative

The ESA_Lab@ concept (non-exchange of funds arrangement between ESA and research entities/universities) has been gathering momentum after two years of testing and build-up. A dozen legally non-binding Memorandums of Collaboration are now in force. The topics vary from the space economy to engineering and technology to exploration and Earth observation

(https://www.esa.int/Enabling_Support/Preparing_for_the_Future/Space_for_Earth/The_ESA_Lab_Initiative).

External Relations

Relations with EU institutions

In March 2019, the EU Council and EU Parliament reached a partial agreement/compromise text on the Space Regulation establishing the Space Programme of the Union and the European Union Agency for the Space Programme under the next EU Multiannual Financial Framework (MFF) 2021–2027.

Since the late summer of 2019, the ESA Executive has been preparing the negotiation of a Financial Framework Partnership Agreement (FFPA) between ESA and the EU based on the EU Financial and Space Regulations and the mandate given to the ESA Director General during the Intermediate Ministerial Meeting (IMM) held in Madrid on 25–26 October 2018. Due to the delay in appointing some members of the European Commission under new President Ursula von der Leyen, the informal negotiations between ESA and the Commission had still not been opened by the end of 2019. Nonetheless, the ESA Director General continued to attend high-level meetings with the European Commission to ensure the seamless functioning of all ongoing ESA-EU cooperation activities.

International Relations

Relations with EU Member States not Members of ESA

ESA has now established formal relations with all the EU Member States not members of ESA. The path towards accession to the Convention was thoroughly reviewed by Council, which adopted on 13 December 2018 a Resolution on industrial policy measures to achieve a successful integration of European States in the framework of ESA. Principles and criteria set out by Council were duly taken into consideration when performing the joint review of the five-year implementation of the European Cooperating State Agreement for Latvia and Lithuania. Both governments applied for Associate membership. A meeting with the Slovenian Minister responsible for space was held in Ljubljana on 9 December to review the status of the Association and negotiate a reinforced Association Agreement, taking into account the lessons learned from the implementation and the above-mentioned improvement measures.

All EU Member States not members of ESA, apart from Malta, attended Space19+ in Seville on 27–28 November and their representatives delivered supportive statements as observers.

Joint reviews (End of PECS Period) for Slovakia, Bulgaria and Cyprus were initiated in 2019.

United States

Following Space19+, ESA and its US counterparts, mainly NASA and NOAA started to discuss the implementation of the Exploration, Earth Observation and Space Weather activities approved by ESA Member States.

In November, ESA and NASA initiated negotiations of a draft Memorandum of Understanding to establish a NASA-ESA partnership on the Lunar Gateway.

After the signature of Statements of Intent on potential ESA contributions to the Mars Sample Return (MSR) campaign in April 2018 and on MSR Campaign Science Benefits in July 2019, NASA submitted to ESA a draft ESA-NASA Responsibilities on MSR leading to the negotiations on the future Memorandum of Understanding.

Bilateral exchanges with NASA on Earth observation continued during the year through the Joint ESA-NASA Programme Planning Group and in ad hoc meetings.

A Letter of Understanding between ESA and NOAA for the preparation of a Space Weather Monitoring System was signed on 19 July.

Canada

The new 10-year Cooperation Agreement was signed on 12 February in Paris at the Embassy of Canada to France. The 40th anniversary of this fruitful cooperation was celebrated on the occasion of the June Council, during which an ESA-CSA Joint Statement was signed.

An ESA-CSA SMILE Letter of Agreement was signed in April.

An Arrangement between ESA and Canada concerning participation by the Government of Canada in the Advanced Research in Telecommunications Systems 4.0 (ARTES 4.0) Programme was approved during the 285th Council meeting held in Seville on 26 November.

Russia

The ExoMars programme continued to be the most prominent cooperation activity for ESA and Roscosmos. ESA, Roscosmos and their respective industrial teams made every possible effort to keep to the July 2020 launch for the ExoMars 2020 mission. An important project milestone was successfully passed with the Qualification Status Checkpoint Review in April. The entire spacecraft composite was sent to Thales Alenia Space in Cannes for thermal vacuum testing.

Cooperation between ESA and Roscosmos in the area of lunar robotics exploration continued.

The launch of the Russian Multipurpose Laboratory Module (MLM) with the European Robotic Arm (ERA) was postponed until the end of 2020.

A programme of joint scientific experiments on board the International Space Station was implemented, which included the ESA-Roscosmos Plasma Kristall-4 experiment and Russian cosmonauts operating the Electro-Magnetic Levitator in the Columbus module.

Soyuz at CSG cooperation proceeded with the launches of three Soyuz-ST vehicles. One of them launched ESA's CHEOPS space telescope.

China

Cooperation continued with China, in particular on the SMILE magnetospheric mission and on the Dragon Earth science joint activities. Several meetings took place to define further cooperation opportunities in space science, in particular in relation to the Einstein probe mission.

Japan

The main highlight for 2019 was when on 14 June, President Hiroshi Yamakawa of JAXA was welcomed to the 282nd meeting of the ESA Council held at ESA's Operations Centre. The JAXA President delivered a presentation highlighting 40 years of cooperation between ESA and JAXA.

ESA and JAXA continued to focus their cooperation on Earth science and space exploration, with the joint EarthCARE mission, the joint BepiColombo mission, the ESA contribution to the JAXA MMX and XRISM X-Ray missions, and joint work on international the habitation module (I-Hab) of the Gateway.

Africa and the Middle East

In January, ESA and the South African National Space Agency (SANSA) held a workshop to agree on short- and medium-term collaborative activities.

Together with the German Aerospace Center (DLR), preparatory activities were carried out with Namibia to launch the DesertSAR airborne campaign in preparation of the Earth Explorer BIOMAS mission.

In September, ESA supported the first UAE astronaut mission on the ISS via the definition and implementation of the selected UAE experiments, provision of astronaut training, support to related education and outreach activities and in the establishment of UAE Operations Support Centre.

India

ESA and ISRO signed in April a Copernicus Space Component Technical Operating Arrangement (TOA) following the signature in 2018 of the Administrative Agreement between EU and the Indian Department of Space to ensure mutual access to the Sentinels satellite data and to the Indian Earth observation satellite data.

Australia

The existing treaty between ESA and the Federal Government of Australia to host the New Norcia deep space tracking station in Western Australia will now see a third antenna built on site, with maintenance and operational support provided by Australia's national science agency CSIRO since mid-2019.

Latin America

A Letter of Intent between ESA and the Planetarium of Buenos Aires was signed on 9 April to enable Argentinean teachers to take part in the CESAR teacher-training programme. ESA organised, together with CONAE and the Planetarium of Buenos Aires, a CESAR teacher-training course on 24 October, with around 120 participants attending. A space day was held on 9 April that was attended by more than 500 scholars.

A Letter of Intent was signed in November with CONAE and the National Directorate for the Antarctic to set up a joint project to use Argentinean Antarctic bases as a ground-based analogue for human space exploration and demonstrate medical equipment.

Following the signature of the ESA-AEB Implementing Arrangement for the setting up and use of telemetry and tracking facilities on Brazilian territory, all ESA family launchers are being tracked from Natal. A Copernicus Space Component Technical Operating Arrangement between ESA, AEB and the National Institute for Space Research (INPE) was signed on 14 March. During the summer, ESA

supported Brazil by providing Sentinel 1 data to monitor the oil spill along the North-East Brazilian coast.

A Copernicus Space Component Technical Operating Arrangement between ESA and the University of Chile was signed on 20 August.

The Mexican Space Agency is participating in the ESA TransparentSea project in conjunction with the Agrifood and Fisheries Information System.

Iceland

Following an invitation from Iceland's Ministry of Foreign Affairs, an ESA delegation met several public and private institutions in Reykjavik on 14 February. The Ministry of Foreign Affairs expressed a desire to begin negotiating a Cooperation Agreement with ESA.

United Nations

ESA participated as an observer in the 56th session of the Scientific and Technical Subcommittee, the 58th session of the Legal Subcommittee and the 62nd session of the Committee on the Peaceful Uses of Outer Space. ESA continued to support the United Nations Office of Outer Space Affairs, in particular for the implementation of two Memorandums of Understanding addressing cooperation on Sustainable Development Goals, the use of the ESA Large Diameter Centrifuge facility, and the issue of space debris as identified in a Joint Statement signed in May 2019.

Education

With dedicated expert support provided by nearly all ESA directorates, the Education Office delivered a wide portfolio of opportunities and activities to meet the needs of educational stakeholders and communities in the ESA Member States.

The STEM Programme, targeting the primary and secondary education levels, has reached an outstanding level of maturity. This stems from the consolidation of the quality status and the effectiveness of the existing ESEROs (European Space Education Resource Offices) in the respective countries; from the development of new educational classroom resources and projects and from a virtuous cycle of structured collaboration and exchange between ESA and the ESEROs (the network dimension). In 2019, the ESERO network further expanded with the establishment of ESERO-Italy; the e-technology lab portfolio, located at the ESA Education Training Centre at ESEC in Belgium, operated as a permanent teacher training facility; and the school projects portfolio (Astro Pi, Climate Detectives, Moon Camp, CanSat, Mission X) was significantly improved.

The ESA Academy, the programme for university students, provided a unique, academically rigorous and professionally solid learning experience to hundreds of university students. The Training and Learning Programme, sharing ESA expertise with university students through dedicated training sessions at the ESEC-based Training and Learning Facility and its educational Concurrent Design Facility, offered a suite of one-week training sessions in a wider range of space-related disciplines. The educational CubeSat programme proposed more frequent student participation opportunities through: CubeSat-related trainings; technical support to CubeSat university student teams at the ESEC CubeSat Support Facility and with the initiation of the third cycle of the *Fly Your Satellite!* Programme. Among other hands-on projects, the Education Office continued with *Fly a Rocket!*, a rocketry programme for Bachelor's students run in collaboration with NAROM, the Andoya Space Centre, and the Norwegian Space Agency; as well as *Orbit Your Thesis!*, in collaboration with the Directorate of

Human Spaceflight and Robotic Exploration, making use of the ICE Cubes commercial facility to host student experiments on board the International Space Station.

The 2019 consolidated participation statistics show that overall, in close collaboration with the ESEROs and both national and European institutional partners and universities, 343 743 primary/secondary teachers were reached (56 004 trained), and 1 885 045 school pupils (1 750 621 through curriculum-relevant activities). Moreover, 2290 university students benefitted from the transfer of ESA expertise through: the ESA Academy hands-on and training opportunities (1216); scholarships and sponsored participation to selected conferences and summer schools (277); Earth Observation training and capacity-building courses (379) and Science lectures (418).

Corporate Development

Knowledge Management

Over more than 50 years of operations, ESA has generated technical, scientific and managerial knowledge that has allowed the European space industry to grow and become competitive in the global marketplace and has fostered scientific progress and technological development in the space domain. This collective knowledge is a key asset for the organisation, for the European space community and for European citizens as a whole.

A comprehensive Agency-wide knowledge management framework was established at ESA in 2017 with the entry into force of the ESA Knowledge Management Policy promoted by the Director General. In the ensuing years, by learning from success pilot activities that have contributed to a better understanding of the needs of the whole ESA community, knowledge management action has been streamlined in two top-priority areas: knowledge from people and lessons learned from projects. The provision of library services across the different ESA sites and activities exploiting knowledge from information and data (i.e. artificial intelligence) support the actions on the two priority areas. In particular, during 2019 the following results have been achieved:

- A process for the retention of core knowledge from key experts leaving the organisation or changing responsibilities has been validated in different areas of the Agency, with the application of modern and effective methods and tools ensuring the capture and accessibility of the retained knowledge.
- The deployment of an enhanced Expert Directory supporting the identification and accessibility of key expertise and knowledge in the Agency has been gradually implemented across the organisation.
- A process for the systematic capture of lessons learned during all the phases of a project has been put in place with the availability of several techniques of capture and the availability of results in a centralised portal.
- A web portal solution to disseminate ESA's knowledge, considering ESA standards on security, data protection and privacy (Knowledge Management Toolkit) has been made available for internal and external access. Examples are the Nebula library¹, and the portal showing how ESA supports the UN Sustainable Development Goals²).
- A free mobile application and Microsoft Word plug-in to provide a glossary³ of terms used in all European space activities have been released and are currently used by the whole European space community.

¹ <https://nebula.esa.int/>

² <https://sdg.esa.int/>

³ https://www.esa.int/About_Us/Business_with_ESA/Small_and_Medium_Sized_Enterprises/Free_app_provides_glossary_of_terms_used_in_all_European_space_activities

Last but not least, following a decision taken by Ministers at the Space19+ Council meeting at ministerial level in Seville, knowledge management is now among the priorities of the ESA Basic Activities preparing the future, and a solid plan of activities for the period 2020–2022 has been approved for implementation.

Corporate Social Responsibility

ESA aims to become a more and more sustainable and responsible organisation, being a role model for the space community in Europe and the world. For more than 10 years, the Agency has been implementing actions aimed at reducing its environmental footprint, increasing the sustainability of its activities and enhancing the contribution of its programmes and missions towards tackling global challenges such as climate change. The last ESA Corporate Social Responsibility Report providing details of the Agency's actions in this area in 2017–2019 was issued in 2019 and is available online⁴.

Besides these activities, and in order to better focus the action on key priorities, an intensive dialogue with the Agency's key stakeholders (representatives of Member States, industry, the scientific community, national space agencies, ESA staff and on-site contractors) was conducted in 2019, following the approach set out in the ISO 26000:2010 standard. The dialogue allowed the identification of the main aspects that ESA should work on in order to improve its contribution to the sustainable development of society.

Based on the results, the environmental impact of ESA's activities and global environmental challenges was considered the aspect of highest importance. The Agency is therefore focussing on actions in the following priority areas:

- Reducing the environmental impact of space missions;
- Reducing the environmental impact of sites and infrastructure; and
- Contributing to environmental monitoring and protection.

Whereas the second area falls completely within the remit of the ESA Director General, a close dialogue with the European space industry and with ESA Member States has been initiated to make progress in the first area (space missions) and in the third one also.

In particular, thanks to extensive work with Large System Integrators, a common set of principles and objectives have been agreed aimed at reducing the environmental impact of space missions and achieving a more responsible industrial supply chain.

⁴ [http://www.esa.int/About_Us/ESA_Publications/SP-1338 ESA Corporate Responsibility and Sustainability 2017 19 Report](http://www.esa.int/About_Us/ESA_Publications/SP-1338_ESA_Corporate_Responsibility_and_Sustainability_2017_19_Report)

Advisory Bodies

Advisory Groups to the ESA Delegate Bodies

1. Technology Harmonisation Advisory Group (THAG)

The purpose of the Technology Harmonisation Advisory Group (THAG) is to advise the IPC on the harmonisation of European programmes in the technological research field and the preference to be given to technology developed within the Member States.

2. Technology Advisory Working Group (TA-WG)

The purpose of the Technology Advisory Working Group (TA-WG) is to advise the IPC on the Agency's technology policy and on the content of the Agency's Technology Programmes.

3. PB-HME advisory group: Exploration and Utilisation Board (EUB)

The HME-EUB is established by the Human Spaceflight, Microgravity and Exploration Programme Board (PB-HME) to support PB-HME, in particular on matters related to science and research, including their technology aspects, and the utilisation aspects of the Agency programmes monitored by PB-HME. The main purpose of HME-EUB is to support the decision-making process at PB-HME in these areas.

4. PB-EO advisory group: Data Operations Scientific and Technical Advisory Group (DOSTAG)

The Earth Observation Programme Board has delegated to DOSTAG various functions limited to data, operations, scientific and technical aspects, for example reviewing the status and risks of all ESA Earth Observation programmes, including space, ground and exploitation segment aspects.

5. SEC advisory group: ESA INFOSEC Panel

The ESA Security Committee (SEC) has delegated to the ESA INFOSEC Panel all matters related to information and communication security.

In this capacity, the ESA INFOSEC Panel supports the ESA Security Office in its function as Security Accreditation Authority (SAA) and provides advice on an effective security programme for ESA in order to ensure the confidentiality, integrity and availability of classified information stored, processed and transmitted.

6. JCB Advisory Committee on 5G (5JAC)

The JCB 5G Advisory Committee (5JAC) is the advisory body to the JCB on matters related to the 5th Generation (5G) of communication networks.

The 5JAC mandate is currently under review, with the objective of aligning it with the ARTES 4.0 Strategic Programme Line "Space for 5G", which Member States endorsed at Space19+ in November 2019. Its terms of reference are currently being revised.

7. Advisory Committee on Space Systems for Safety and Security (4S) (formerly on Governmental Secure Satcom)

JCB Governmental Secure Satcom Advisory Committee is the advisory body to the JCB on matters related to ESA's Govsatcom-related programmes and activities.

The advisory committee's mandate is currently under review, with the objective of aligning it with the ARTES 4.0 Strategic Programme Line "Space Systems for Safety and Security" (4S), which Member States endorsed at Space19+ in November 2019.

8. Optical Communication - ScyLight Advisory Committee (SCOTT)

The JCB SCyLight Optical communication advisory committee (SCOTT) is the advisory body to the JCB on matters related to the activities carried out under the ARTES 4.0 Strategic Programme Line "Optical Communication – ScyLight". It was proposed to establish the advisory committee following Member States' endorsement of the Strategic Programme Line at Space19+ in November 2019. The advisory committee's terms of reference are currently under definition.

Advisory Groups to the ESA Director General

Science advisory committees

The role of the science advisory committees is to advise on science and scientific activities taking into account the recommendations of the science working groups (see below).

The science advisory committees are the Space Science Advisory Committee (SSAC), the Advisory Committee for Earth Observation (ACEO), the GNSS Science Advisory Committee (GSAC) and the Human Spaceflight and Exploration Science Advisory Committee (HESAC).

Space Science Advisory Committee (SSAC)

The Space Science Advisory Committee (SSAC) is the senior advisory body to the Director of Science (D/SCI), acting as the Director General's representative on all matters concerning space science included in the mandatory science programme of ESA. It is the main interpreter of the views and needs of the European scientific community as regards access to space experimentation and data exploitation in the mandatory science programmes.

Advisory Committee for Earth Observation (ACEO)

The Advisory Committee for Earth Observation (ACEO) is the senior advisory body to the Director of Earth Observation (D/EOP), acting as the Director General's representative on all matters concerning Earth Sciences included in the Earth Observation programmes of ESA. It is the main interpreter of the views and needs of the European scientific community as regards access to space experimentation and data exploitation in the Earth science programme.

ACEO also has the highest-level advisory capacity, through the Director of Earth Observation (D/EOP) on matters to be treated at the level of the Programme Board for Earth Observation (PB-EO), which may also request advice on particular issues of a scientific nature.

The transition from the existing ESAC (Earth Science Advisory Committee) to the newly-created ACEO was discussed during the 173rd meeting of PB-EO in February 2018, ref. ESAPB-EO(2018)10, including a description of the terms of reference of ACEO.

GNSS Science Advisory Committee (GSAC)

The Global Navigation Satellite System (GNSS) Science Advisory Committee (GSAC) is the senior advisory body to the Director of Navigation (D/NAV) and the Director of Science (D/SCI), acting as the Director General's representatives on all matters concerning scientific exploitation of navigation programmes. It is the main interpreter of the views and needs of the European scientific community as regards uses of the GNSS Programme for research in Earth, navigational and physical sciences.

Human Spaceflight and Exploration Science Advisory Committee (HESAC)

The Human spaceflight and Exploration Science Advisory Committee (HESAC) is the senior advisory body to the Directors of Human and Robotic Exploration (D/HRE) and Science (D/SCI), acting as the Director General's representatives on all matters concerning scientific exploitation of human spaceflight, microgravity and exploration programmes. It is the main interpreter of the views and needs of the European scientific community as regards access to space experimentation and data exploitation in the human spaceflight, microgravity and exploration programmes.

Science working groups

Working groups report to the advisory committees listed above on matters concerning their specific science areas. In order to provide a single set of science disciplinary working groups, avoiding duplication or conflicts of advice, the following are established to provide advice on their disciplines across the different science advisory committees.

The working groups support the work of all the science advisory committees and Directorates as appropriate. Any Directorate can request appropriate advice from any of the Working Groups.

Astronomy Working Group (AWG)

The Astronomy Working Group (AWG) advises on all science matters related to astronomy. It reports mainly to SSAC but also to HESAC and ACEO (formerly ESAC).

Physical Sciences Working Group (PSWG)

The Physical Sciences Working Group (PSWG) advises on all aspects of material sciences, fundamental physics and experiments in microgravity conditions. It reports mainly to HESAC and SSAC, but also to ACEO (formerly ESAC) and GSAC.

Solar System Exploration Working Group (SSEWG)

The Solar System Exploration Working Group (SSEWG) advises on all science matters related to solar system science and exploration, excluding Earth sciences. It reports mainly to SSAC but also to HESAC and ACEO (formerly ESAC).

Accounts

Notes to the Financial Statements

As required by the ESA Financial Regulations, the Agency's annual Financial Statements are prepared in accordance with International Public Sector Accounting Standards (IPSAS).

The Financial Information presented below is extracted from the audited ESA Financial Statements for the financial year 2019, and consists of the following components:

- Statement of Financial Position;
- Statement of Financial Performance;
- Statement of Changes in Net Assets/Equity;
- Statement of Cash Flows.

The Net Surplus for 2019 amounts to €365.8m, after capitalisation of Assets Under Construction and work in progress, as compared to a Net Surplus of €503.3m in 2018. The decrease in the 2019 result by €137.5m compared to 2018 is mainly due to the decreased result of ESA programmes of €424.7m, partially offset by the increased result for pensions of €280.7m.

The 2019 net surplus breaks down into:

- €326.2m net surplus for ESA (€750.9m net surplus in 2018)
- €32.0m net surplus for Pensions (€248.7m net deficit in 2018)
- €7.7m net surplus for Social Security (€1.1m net surplus in 2018)

ESA's Financial Statements are audited by the Audit Commission of the European Space Agency in accordance with the International Standards on Auditing. In the opinion of the Audit Commission, the 2019 Financial Statements give a true and fair view of the Financial Position of the European Space Agency as at 31 December 2019, and of its Financial Performance and its Cash Flows for the 12-month period ended 31 December 2019. Moreover, the Audit Commission reported that nothing came to its attention which would indicate that the transactions of ESA have not been made, in all significant respects, in accordance with the Regulations of the European Space Agency.

The audited Financial Statements and the Independent Auditor's report were submitted to Council in accordance with the ESA Financial Regulations, which were approved by the Council Meeting on 24 June 2020.

Statement of financial position

Statement of Financial Position

	Notes	31 December 2019 million €	31 December 2018 million €
Total Assets		13,112.9	12,834.3
Current assets		3,098.7	3,030.3
Cash and cash equivalents	3.1	1,302.7	923.6
Other financial assets		0.1	0.0
Recoverables from non-exchange transactions	3.2	412.2	476.6
Receivables from exchange transactions	3.3	801.2	1,193.8
Inventories	3.4	582.5	436.3
Non-current assets		10,014.2	9,804.0
Recoverables from non-exchange transactions	3.2	3.6	3.8
Receivables from exchange transactions	3.3	407.5	326.3
Financial assets	3.5	859.2	721.6
Property, plant and equipment	3.6	8,451.2	8,483.8
Intangible assets	3.6	292.7	268.5
Total Liabilities		11,783.1	10,248.9
Current liabilities		2,876.7	3,127.3
Payables under exchange transactions		2,302.4	2,497.7
Current payables	3.7	1,183.9	1,548.4
Pre-financing from exchange transactions	3.8	1,061.2	907.9
Deferred income	3.9	57.3	41.4
Contributions and other payables		358.5	424.0
Prepaid contributions & other amounts payable to Member/Participating States	3.10	277.2	190.8
Pre-financing from non-exchange transactions	3.11	5.2	6.1
Other financial liabilities	3.13	0.0	200.3
Other payables	3.12	76.1	26.8
Provision for employee benefits	3.14	207.0	190.6
Provisions for risks and charges	3.15	8.8	15.0
Non-current liabilities		8,906.4	7,121.6
Provision for employee benefits	3.14	8,603.7	6,890.4
Other financial liabilities	3.13	0.0	0.1
Other payables	3.12	302.1	172.0
Pre-financing from non-exchange transactions	3.11	0.6	0.0
Pre-financing from exchange transactions	3.8	0.0	59.1
Net Assets		1,329.8	2,585.4
Reserves		4,371.0	5,240.4
Amounts to be called from Member States	3.16	-3,407.0	-3,158.3
Net Surplus/(Deficit) for the period	II	365.8	503.3
Net Assets		1,329.8	2,585.4

Statement of financial performance

Statement of Financial Performance

	Notes	2019 million €	2018 million €
Operating Revenues	4.1	5,728.8	5,306.3
Contributions from Member/Participating States		4,043.6	3,818.2
Other contributions & grants		91.3	98.3
Revenues from exchange transactions		1,078.8	1,104.1
Staff contributions		272.4	45.0
Other operating income		242.6	240.7
Operating Expenses	4.1	-5,437.9	-4,788.2
Purchases		-2,453.2	-2,182.7
External services		-1,063.9	-988.0
Staff expenses		-681.4	-624.8
Other operating expenses		-242.9	-235.4
Expenses from non-exchange transactions		-99.0	-97.1
Depreciation and amortization		-849.9	-624.6
Impairment		7.9	2.2
Write-downs inventories and fixed assets		-55.5	-37.8
Net Surplus / (Deficit) from Operating activities		290.8	518.1
Financial activities	4.2	75.0	-14.8
Financial revenue		79.4	-6.4
Financial expenses		-4.4	-8.4
Net Surplus / (Deficit) for the period		365.8	503.3

Statement of changes in net assets /equity

Statement of Changes in Net Assets/Equity

		31 December 2018 million €	Allocation of audited 2018 surplus/ (deficit) million €	Movements 2019 million €	31 December 2019 million €
RESERVES	A	5,240.4	752.0	-1,621.4	4,371.0
Member States Capital and Reserves		-980.8	-574.0	-1,622.1	-3,176.9
Revaluation Reserve	(*)	1,826.0	-256.6	-1.1	1,568.3
Currency Exchange Gains Reserve		3.0			3.0
Sundry Reserves		0.1			0.1
Other Reserves		-2,809.9	-317.4	-1,621.0	-4,748.3
Social security reserve		26.8	1.1	0.7	28.6
Buffer Fund fair value reserve		1.0		12.1	13.1
Reserve Barter		35.4	-318.5		-283.1
Actuarial gains and losses		-2,873.1		-1,633.8	-4,506.9
Accumulated Surplus		6,221.1	1,326.0	0.7	7,547.8
Accumulated Surplus ESA programmes		6,067.6	1,296.1	0.7	7,364.4
Accumulated Surplus Third Party programmes		139.1	27.0		166.1
Accumulated Surplus PECS		14.4	3.0		17.4
AMOUNTS TO BE CALLED FROM MEMBER STATES	B	-3,158.3	-248.7	0.0	-3,407.0
Member States pension scheme receivable		-3,158.3	-248.7		-3,407.0
Accumulated Deficit					
Net Surplus / (Deficit) for the Period	C	503.3	-503.3	365.8	365.8
TOTAL NET ASSETS	A+B+C	2,585.4	0.0	-1,255.6	1,329.8

(*) First-Time IPSAS adoption

Statement of Changes in Net Assets/Equity

		31 December 2017 Restated	Allocation of audit 2017 surplus/ (deficit)	Allocation of restated 2017 IPSAS 39 surplus	Allocation of restated 2017 Barter deficit	Movements 2018	31 December 2018
		million €	million €		million €	million €	million €
RESERVES	A	4,762.3	732.8		-154.7	-100.0	5,240.4
Member States Capital and Reserves		-288.6	-375.7		-154.7	-161.8	-980.8
Revaluation Reserve (*)		2,262.6	-374.9			-61.7	1,826.0
Currency Exchange Gains Reserve		3.4	-0.4				3.0
Sundry Reserves		0.1					0.1
Other Reserves		-2,554.7	-0.4		-154.7	-100.1	-2,809.9
<i>Social security reserve</i>		27.5	-0.4			-0.3	26.8
<i>Buffer Fund fair value reserve</i>		6.8				-5.8	1.0
<i>Reserve Barter</i>		190.1			-154.7		35.4
<i>Actuarial gains and losses</i>		-2,779.1				-94.0	-2,873.1
Accumulated Surplus		5,050.9	1,108.5			61.7	6,221.1
<i>Accumulated Surplus ESA programmes</i>		4,920.1	1,100.9			46.6	6,067.6
<i>Accumulated Surplus Third Party programmes</i>		117.9	6.1			15.1	139.1
<i>Accumulated Surplus PECS</i>		12.9	1.5				14.4
AMOUNTS TO BE CALLED FROM MEMBER STATES	B	-2,941.5	-355.8	138.9	0.0	0.0	-3,158.3
Member States pension scheme receivable		-2,941.5	-355.8	138.9			-3,158.3
Accumulated Deficit							
Net Surplus / (Deficit) for the Period	C	361.2	-377.0	-138.9	154.7	503.3	503.3
TOTAL NET ASSETS	A+B+C	2,182.0				403.3	2,585.4

(*) First-time IPSAS adoption

Statement of cash flow

Statement of Cash Flows

	2019	2018
	million €	million €
Net Cash Flows	379.1	246.3
Net Cash Flows from Operating Activities	1,520.7	2,027.0
Surplus/(deficit) from ordinary activities	365.8	503.3
<i>Reclassification to investing activities</i>	-16.3	-11.5
<i>Non-cash movements</i>	606.1	904.3
Depreciation & impairment	842.0	622.4
Provisions & Accruals	-6.0	-1.1
Write-downs / Adjustments inventories and fixed assets	55.0	262.2
Revaluation of financial assets at fair value	-60.6	20.8
Past service costs	-224.3	0.0
<i>Decrease (Increase) in Deferrals/accruals of past or future operating cash receipts or payments</i>	565.0	630.9
Decrease/(increase) in inventories	-141.4	-84.5
Decrease/(increase) in recoverables from Member States	65.5	-5.1
Decrease/(increase) in other receivables	-88.5	244.5
Decrease/(increase) in other current assets	399.2	-199.3
Increase/(decrease) in payable to Member/Participating States	85.6	-9.8
Increase/(decrease) in pre-financing	93.8	69.7
Increase/(decrease) in other payables	-169.3	332.2
Increase/(decrease) in other financial liabilities	0.0	0.0
Increase/(decrease) in provisions for employee benefits	320.2	283.2
Net Cash Flows from Investing Activities	-941.6	-1,730.7
Acquisition of fixed assets	-893.4	-1,688.8
Proceeds from sales of fixed assets	0.0	0.0
Purchases of financial investments (portfolio)	-149.9	-148.5
Proceeds on disposal of financial investments (portfolio)	93.6	98.1
Interests received on financial assets	8.0	8.5
Net Cash Flows from Financing Activities	-200.0	-50.0
Proceeds from borrowings	200.0	200.0
Repayment of borrowings	-400.0	-250.0
Net increase/(decrease) in cash and cash equivalents	379.1	246.3
Cash and cash equivalents at the beginning of the year	923.6	677.3
Cash and cash equivalents at year-end	1,302.7	923.6

ANNEX

Annex 1 – Agreements signed in 2019

- ✓ The Memorandum of Understanding between the European Space Agency and the United Nations on the Use of the ESTEC Large Diameter Centrifuge was signed on 14 February 2019 in Paris by Mr Johann-Dietrich Wörner, ESA Director General, and on 15 February 2019 in Vienna by Simonetta di Pippo, Director of UNCOPUOS. **(LEG/491)**
- ✓ The Memorandum of Understanding between the European Space Agency and the United Nations on Cooperation for the Achievement of the United Nations Sustainable Development Goals was signed on 24 May 2019 in Paris by Mr Johann-Dietrich Wörner, ESA Director General, and Simonetta di Pippo, Director of UNCOPUOS. **(LEG/500)**
- ✓ The Implementing Arrangement between ESA and the National Space Science Centre of the Chinese Academy of Sciences concerning the Einstein Probe Mission was signed on 7 May 2019 in Villanueva de la Cañada by Mr Günther Hasinger, ESA Director of Science, and Mr Wang Chi, Director General of NSSC. **(LEG/492)**
- ✓ The Letter of Agreement regarding the Arrangement for the Solar Wind Magnetosphere Ionosphere Link Explorer (SMILE) Mission was signed on 10 April 2019 in Colorado Springs by Mr Johann-Dietrich Wörner, ESA Director General, and Mr Sylvain Laporte, President of the Canadian Space Agency. **(LEG/493)**
- ✓ The Agreement Modifying the Agreement between Romania and the European Space Agency concerning the Accession of Romania to the Convention for the Establishment of a European Space Agency and Related Terms and Conditions was signed on 14 May 2019 in Paris by Mr Johann-Dietrich Wörner, ESA Director General, and Mr Marius-Ioan Piso, Executive Director of the Romanian Space Agency. **(LEG/494)**
- ✓ The Project Arrangement for ESA Assistance in the Management and Implementation of the Greek National Satellite Space Project was signed on 20 March 2019 in Marseille by Mr Johann-Dietrich Wörner, ESA Director General, by Mr Nikkos Pappas, representing the Greek Minister of Digital Policy, Telecommunications and Media and by Mr Yiannis Dragasakis, representing the Greek Minister of Economy and Development. **(LEG/495)**
- ✓ The Cooperation Agreement between the European Space Agency and the Government of Canada was signed on 14 February 2019 in Paris by Mr. Johann-Dietrich Wörner, ESA Director General, and Mr Sylvain Laporte, President of the Canadian Space Agency, for the Government of Canada. **(LEG/496)**
- ✓ The Contribution Agreement between the European Space Agency and the European Union, represented by the European Commission, on Space Technology Activities under the Horizon 2020 Programme was signed on 16 April 2019 in Brussels by Mr Johann-Dietrich Wörner, ESA Director General, and by Mr Timo Pesonen, Director General for Internal Market, Industry, Entrepreneurship and SMEs. **(LEG/497)**

- ✓ The Agreement between the European Space Agency and the United Kingdom Space Agency concerning the SMILE Mission was signed on 19 June 2019 in Paris by Mr Günther Hasinger, ESA Director for Science, and Mr Chris Lee, Head of Science Programmes at UKSA. **(LEG/498)**
- ✓ The Implementing Arrangement between the European Space Agency (ESA) and the Facility for Antiproton and Ion Research in Europe GmbH (FAIR) on their cooperation for the ESA-FAIR Radiation Summer School was signed in Paris on 17 May 2019 by Mr Johann-Dietrich Wörner, ESA Director General, Mr Jörg Blaurock, FAIR Technical Managing Director, Mr Paolo Guberlino, FAIR Scientific Managing Director, and Mrs Ursula Weyrich, FAIR Administrative Managing Director. **(LEG/501)**
- ✓ The Agreement between the European Space Agency and the Government of the Republic of Portugal concerning the Provision of Technical Assistance and Expertise to the Azores International Satellite Launch Programme was signed on 21 June 2019 in Vila do Port, Santa Maria, Azores, by Mr Johann-Dietrich Wörner, ESA Director General, and Mr Manuel Heitor, Minister for Science, Technology and Higher Education. **(LEG/502)**
- ✓ The Rider Modifying the Agreement between the European Space Agency and the Government of the Republic of Estonia concerning the Accession of Estonia to the Convention for the Establishment of a European Space Agency and Related Terms and Conditions was signed on 12 November 2019 in Paris by Mr Johann-Dietrich Wörner, ESA Director General, and on 15 November 2019 in Tallinn by the Estonian counterpart. **(LEG/442, add.1)**
- ✓ The Protocol of Extension and Amendment to the Arrangement between the European Space Agency and the Federal Space Agency of Russia concerning cooperation in the conduct of joint experiments on the International Space Station was signed on 26 November 2019 by Johann-Dietrich Wörner, ESA Director General and Dr D. Rogozin, Director General of the State Space Corporation Roscosmos. **(LEG/368, add.1)**
- ✓ The Memorandum of Cooperation between the European Space Agency and the Government of the Grand Duchy of Luxembourg concerning cooperation in the field of space resources was signed on 26 November 2019 in Seville by Mr Johann-Dietrich Wörner, ESA Director General, and Mr Etienne Schneider, Deputy Prime Minister. **(LEG/499)**
- ✓ The Implementing Arrangement between the European Space Agency (ESA) and the European Defence Agency (EDA) on their cooperation on a demonstration project entitled “Autonomous Drone Services - AUDROS” was signed on 19 December 2019 in Brussels by Mr Jorge Domecq, EDA Chief Executive, and on 20 December 2019 in Paris by Mrs Magali Vaissière, ESA Director of Telecommunications and Integrated Applications. **(LEG/503)**
- ✓ The Implementing Arrangement between the European Space Agency (ESA) and the European Defence Agency (EDA) on their cooperation for Artificial Intelligence for Terrain relative Navigation in Unknown Environment (ATENA) Project. **(LEG/504)**

Annex 2 – Patent groups

The following information summarises part of the work carried out by the Patents Group (PG), coordinated by the PG Secretariat, throughout the year 2019. The Secretariat is embedded in the Technology Transfer and Patent Office (TEC-ST) to connect both activities. The legal aspects related to the patents are dealt with by the Legal Department, IPL-LI.

Invention disclosures received in 2019

The Secretariat received and processed 26 new invention disclosures—5 more than in the previous year. 100% of the invention disclosures received have been processed by the Secretariat.

ESA Patents Group decided to protect 8 inventions in its own name. 3 invention disclosures will be published in order to create prior art. The rest will be presented to the Patents Group in 2020 as soon as the inventors provide all the elements required.

The table below lists the inventions for which ESA Patents Group has decided to file a Patent Application.

ESA ref	Title	Decision	Origin of the invention	Ownership
735	Magnetic damping for spacecraft stabilisation after end of life	File European patent application	in-house research	ESA
742	Global Artificial Intelligence-driven Cognitive Network	File International patent application	in-house research	ESA
747	Compact Waveguide Orthomode Transducer	File International patent application	in-house research	ESA
750	Multifocal discrete lens beamforming networks and antennas with controllable front to back aperture ratio	File International patent application	in-house research	ESA
754	Peak and Valley Current Mode Control Using Double Compensation Ramp	File International patent application	in-house research	ESA
756	Dual-Polarisation Four-Way Power Divider	File International patent application	in-house research	ESA
749	Signal and method for fast time ambiguity resolution	File International patent application	in-house research	ESA
752	Multifunction electronic timepiece for navigational / operational aid of an earth-mars space mission	File International patent application	in-house research	ESA

Table 1: List of inventions for which a Patent Application will be filed

Decisions concerning the extension of protection

The Secretariat regularly reviews the patents portfolio (when a milestone is reached, e.g. payment of annuities, extension of protection to other countries or grant of a patent) to decide whether the protection should be continued, validated (in the case of the European Patent) or extended. The table below shows the decisions about the continuation or the extension of protection of 11 inventions and their related patents.

ESA/ PAT	TITLE	TYPE OF DECISION	DECISION
702	A directional Coupler and Method of Manufacturing Thereof	Extension	Extension to US and CA
542	Methods, apparatuses and system for asynchronous spread-spectrum communication	Validation	Validate in FR, UK, ES, IT, DE, NO, CH and NL
597	Beam-forming network for an array antenna and array antenna comprising the same	Validation	Validate in UK, FR, DE, and IT
623	Method and apparatus for transmitting data packets over a transmission channel shared by a plurality of users	Validation	Validate in UK, FR, DE, IT, CH, LU, NL
642	Fluidic filter	Validation	Validate FR, IT, DE and UK
703	Low-overhead compression of time series data	Extension	Extension in US and EP
714	Exothermic Reaction aided Spacecraft Demise during Reentry	Extension	Extension to US
714	Exothermic Reaction aided Spacecraft Demise during Reentry	Extension	Extension to US
584	Data detection method and data detector for signals transmitted over a communication channel with inter-symbol interference	Validation	Validate in GB, DE, FR
682	Imaging spectrometer with reflective grating	Validation	Validate in GB, FR, CH, DE, BE, NL
725	Method and system for processing a gnss signal using homomorphic encryption"	Extension	Extension to US

Table 2 List of decision taken by PG concerning validation and extension of protection

Patents granted in 2019

In 2019, 50 patents were granted improving further industry's and ESA's positions in the related technical fields.

By the end of 2019, 77 patent applications were still under examination by different patent offices around the world. However, a very large part of the portfolio was composed of granted patents (457), which means that the portfolio is relatively strong, as the protection has been endorsed by the patent offices.

A list of patents granted in 2019 in the different jurisdictions is given in the table below:

ESA/PAT	TITLE	COUNTRY OF GRANT
542	Methods, apparatuses and system for asynchronous spread-spectrum communication	FR, DE, UK, CH, IT, ES, NL, NO
569	Array antenna having a radiation pattern with a controlled envelope, and method of manufacturing it	FR, DE, UK
584	Data detection method and data detector for signals transmitted over a communication channel with inter-symbol interference	FR, DE, UK
597	Beam-forming network for an array antenna and array antenna comprising the same	DE, FR, UK, IT
598	Uplink power control, method and apparatus for satellite communications network	IL
610	Reconfigurable RF front end circuit for a multi-beam array fed reflector antenna system	US
623	Asynchronous contention resolution diversity aloha	UK, FR, DE, IT, CH, LU, NL
630	Manufacturing of a metal component or a metal matrix composite component involving contactless induction of high-frequency vibrations	US
635	Foil deployment mechanism	DE, FR, NL, UK
638	Manufacturing of a ceramic article from a metal preform or metal matrix composite preform provided by 3d-printing or 3d-weaving	US
641	Fluidic filter	US, IT, FR, DE, UK
657	Waveguide Branching Switch	US
658	Interferometric Radio Occultation	US
682	Imaging spectrometer with reflective grating	US

Table 3 : list of patents granted in 2019 in the different jurisdictions

Annex 3 - Trademark renewals / extensions registered in 2019

The table below lists the **trademark renewals / extensions registered in 2019**, on behalf of the European Space Agency. By conducting such formalities, the Agency intends to secure exclusive rights in these verbal marks or logos and to create the possibility of protecting them against registration by other parties or unauthorised use.

Trademark	Verbal mark or Logo	Country	Class(es) of Goods and Services	Date of registration
DABYS	Verbal mark	France	9,42	02/05/2019
EGOS		France	9,42	29/08/2019
MICONYS	Verbal mark	France	9,42	29/08/2019
SCOS-2000	Verbal mark	France	9,42	02/05/2019
SIMULUS		France	9,42	29/08/2019
TEVALIS		France	9,42	29/08/2019
VEGA	Verbal mark	France	12	08/04/2019
ARIANE	Verbal mark	Austria	7, 9, 11, 12, 37, 39, 42	25/01/2019
ARIANE	Verbal mark	France	7, 9, 11, 37	22/02/2019
ARIANE	Verbal mark	French Polynesia	7, 9, 11, 37	22/02/2019
ARIANE	Verbal mark	Italy	7, 9, 11, 12, 37, 39, 42	20/05/2019
ARIANE	Verbal mark	Norway	7, 9, 11, 12, 13, 37, 39, 42	16/07/2019
ARIANE	Verbal mark	Spain	11, 12, 39, 42	10/04/2019
ARIANE	Verbal mark	Switzerland	7, 9, 11, 12, 39, 42	21/03/2019

“Class(es) of Goods and Services” means one or more of the groups of products and services set forth in the Nice International Classification, i.e. the International Classification of Goods and Services for the Purposes of the Registration of Marks established by the Nice Agreement of 1957 and available on the WIPO website

(<https://www.wipo.int/treaties/en/classification/nice/>). “WIPO” stands for the World Intellectual Property Organization.

The standard lists of products and services corresponding to Classes 7, 9, 11, 12, 13, 37, 39 and 42 are the ones mentioned, for information, below:

- Class 7 Machines, machine tools, power-operated tools; motors and engines, except for land vehicles; machine coupling and transmission components, except for land vehicles; agricultural implements, other than hand-operated hand tools; incubators for eggs; automatic vending machines.
- Class 9 Scientific, research, navigation, surveying, photographic, cinematographic, audiovisual, optical, weighing, measuring, signalling, detecting, testing, inspecting, life-saving and teaching apparatus and instruments; apparatus and instruments for conducting, switching, transforming, accumulating, regulating or controlling the distribution or use of electricity; apparatus and instruments for recording, transmitting, reproducing or processing sound, images or data; recorded and

downloadable media, computer software, blank digital or analogue recording and storage media; mechanisms for coin-operated apparatus; cash registers, calculating devices; computers and computer peripheral devices; diving suits, divers' masks, ear plugs for divers, nose clips for divers and swimmers, gloves for divers, breathing apparatus for underwater swimming; fire-extinguishing apparatus.

Class 11 Apparatus and installations for lighting, heating, cooling, steam generating, cooking, drying, ventilating, water supply and sanitary purposes.

Class 12 Vehicles; apparatus for locomotion by land, air or water.

Class 37 Construction services; installation and repair services; mining extraction, oil and gas drilling.

Class 39 Transport; packaging and storage of goods; travel arrangement.

Class 42 Scientific and technological services and research and design relating thereto; industrial analysis, industrial research and industrial design services; quality control and authentication services; design and development of computer hardware and software.

Annex 4 Chairs of Council and Delegate Bodies (as of 31/12/2019)

Chair of the Council

Jean-Yves Le Gall (FR)

Vice-Chair

Alice Bunn (UK)

Marc Serres (LU)

Science Programme Committee (SPC)

Stefano Vitale (IT)

Administrative and Finance Committee (AFC)

Christine Klein (DE)

Industrial Policy Committee (IPC)

Christer Nilsson (SE)

International Relations Committee (IRC)

Katrin Merike Nyman-Metcalf (EE)

Security Committee (SEC)

Andrew Standeven (UK)

Programme Board on Satellite Navigation (PB-NAV)

Vaclav Kobera (CZ)

Joint Board on Communication Satellite Programmes (JCB)

Mathias Link (LU)

Earth Observation Programme Board (PB-EO)

Joost Carpay (NL)

Data Operations Scientific and Technical Advisory Group
(PB-EO-DOSTAG)

Thomas Geist (AT)

Programme Board for Human Spaceflight, Microgravity
and Exploration (PB-HME)

Pierre Coquay (BE)

Exploration and Utilisation Board (PB-HME-EUB)

Oliver Botta (CH)

Launchers Programme Board (PB-LAU)

Roberto Trigo (ES)

Programme Board for Space Situational Awareness
(PB-SSA)

Johan Köhler (SE)

Annex 5: National Delegations to Council (as of 31/12/2019)

AUSTRIA

Klaus Pseiner
Andreas Geisler
Margit Mischkulnig

BELGIUM

Frank Monteny
Pierre Coquay

CZECH REPUBLIC

Václav Kobera

DENMARK

Peter Sloth
Gorm Kofoed Petersen

ESTONIA

Madis Võõras
Paul Liias

FINLAND

Kimmo Kanto
Maija Lönnqvist
Pauli Stigell

FRANCE

Jean-Pascal Le Franc
Lionel Suchet
Christophe Venet

GERMANY

Walther Pelzer
Gerd Kraft
Klaus Steinberg
Max Croymann

GREECE

Giorgios Synnefakis
Athanasios Paliatsos
Georgios Mantzouris

HUNGARY

Orsolya Ferencz
Ferenc Horvai

IRELAND

Ciara Bartley
Tony McDonald

ITALY

Giorgio Saccoccia
Maria-Christina Falvella
Marco Maria Cerbo

LUXEMBOURG

Marc Serres
Mathias Link

NETHERLANDS

Eelco van der Eijk
Harm van de Wetering

NORWAY

Christian Hauglie-Hanssen
Geir Hovmork
Knut Myrvang

POLAND

Robert Nowicki
Julita Wilczek

PORTUGAL

Chiara Manfletti
Luis Serina

ROMANIA

Marius-Ioan Piso
Oana Neagu

SPAIN

Javier Ponce
Juan Carlos Cortés Pulido
Jorge Lomba

SWEDEN

Anna Rathsmann
Thorwald Larsson
Magnus Härviden

SWITZERLAND

Renato Krpoun
Raphaël von Roten
Urs Frei

UNITED KINGDOM

Maria Cody
Graham Turnock
Mark Bacon

CANADA

Sylvain Laporte
Marie-Claude Guérard
Jean-Pierre Arseneault

SLOVENIA

Tanja Permozer